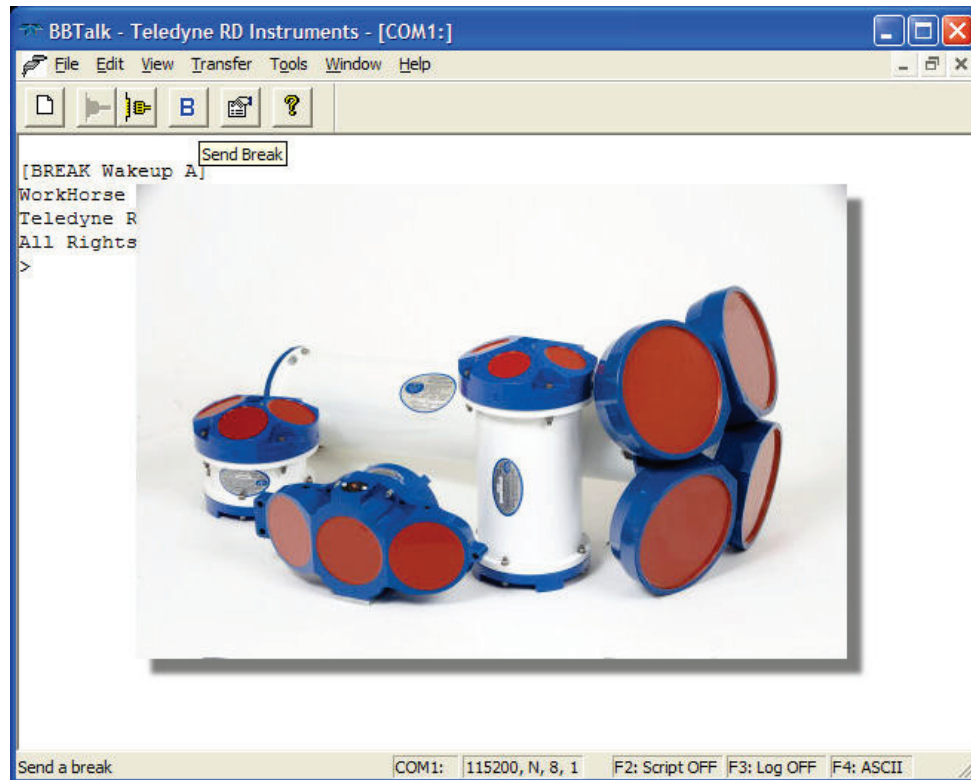


RDI Tools User's Guide



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TELEDYNE
RD INSTRUMENTS

A Teledyne Technologies Company

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RDI Tools User's Guide

1 Introduction

This guide is an overview on using the *RDI Tools* utility software provided with your system. Use *BBTalk* to communicate with the ADCP.



NOTE. This guide covers version 3.08 or higher.

1.1 How to Contact Teledyne RD Instruments

If you have technical issues or questions involving a specific application or deployment with your instrument, contact our Field Service group:

Teledyne RD Instruments

14020 Stowe Drive
Poway, California 92064

Phone +1 (858) 842-2600

FAX +1 (858) 842-2822

Sales – rdisales@teledyne.com

Field Service – rdifs@teledyne.com

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Sales – rdie@teledyne.com

Field Service – rdiefs@teledyne.com

Client Services Administration – rdicsadmin@teledyne.com

Web: <http://www.rdinstruments.com>

24 Hour Emergency Support +1 (858) 842-2700

1.2 System Requirements

TRDI software requires a Windows® compatible computer with the following specifications:

- Windows XP®, Windows 2000®, or Vista
- Pentium III 600 MHz class PC (higher recommended)
- 64 megabytes of RAM (128 MB RAM recommended)
- 50 MB Free Disk Space plus space for data files (A large, fast hard disk is recommended)
- One Serial Port (two or more High Speed UART Serial Port recommended)
- Minimum display resolution of 1024 x 768, 256 color (higher recommended)
- CD-ROM Drive
- Mouse or other pointing device

1.3 Software Installation

To install the RDI Tools software, do the following.

- a. Insert the compact disc into your CD-ROM drive and then follow the browser instructions on your screen. If the browser does not appear, complete Steps “b” through “d.”
- b. Click the **Start** button, and then click **Run**.
- c. Type **<drive>:launch**. For example, if your CD-ROM drive is drive D, type **d:launch**.
- d. Follow the browser instructions on your screen.

Once, installed, you will have several shortcuts added to your Windows® **Start** menu. The **BBTalk** shortcut will start *BBTalk*. **BBLIST** will start the DOS *BBLIST* program.



NOTE. The default directory for *BBTalk* installation is *C:\program files\RD Instruments\RDI Tools*.

2 Using BBTalk

BBTalk is a dumb terminal emulator program. This Windows® compatible program can capture raw data files and help troubleshoot configuration problems. You can use *BBTalk* for serial or parallel communications in either an ASCII or BINARY mode. A binary-to-hexadecimal conversion feature lets you view and record the binary output data in a hexadecimal format. A LOG feature lets you record data to a disk file.

2.1 BBTalk Short-Cut Keys

Use the following keys to quickly start tasks.

Table 1: BBTalk Short-Cut Keys

Key	Description
End	Sends a BREAK to the ADCP (wakes up the ADCP).
F1	Brings up the BBTalk Help file.
F2	Runs/Stops a script file.
F3	Starts/Stops recording data to a Log File.
F4	Switches between ASCII and Hex display.
F5	Brings up the Communications Properties window.
F8	This will connect the ADCP to the Com Port.
F9	This will disconnect the ADCP from the Com Port.
Ctrl+B	Sends binary data in a HEX format. For example, typing "4142" and pressing enter sends the characters "AB" to the ADCP.
Ctrl+C	Copies content of the selected text in the communication windows.
Ctrl+N	Connects to a new COM Port.
Ctrl+PageDown	Recover the ADCP's internal recorder data.
Ctrl+T	Sends PC time to ADCP (synchronizes PC and ADCP clocks).
Ctrl+V	Sends clipboard content to the ADCP.
Up	Shows the Command History window.



NOTE. Using the F4 key to switch between ASCII and Hex will also determine what type data will be recorded to the Log file. For example, pressing F4 to switch to the Hex mode will cause Hex data to be recorded to the Log file.

2.2 Communication Parameters

Before you can establish communications with the ADCP, you must configure *BBTalk*.

- a. At the **Connect To** screen, select the ADCP type (**WorkHorse**, **BroadBand**, **NarrowBand**, **Channel Master**, **NEMO**, **AccQmin**, **DVS**, or **RiverRay**) from the list.
- b. Select the **COM Port** the ADCP is connected to. Click **Next**.

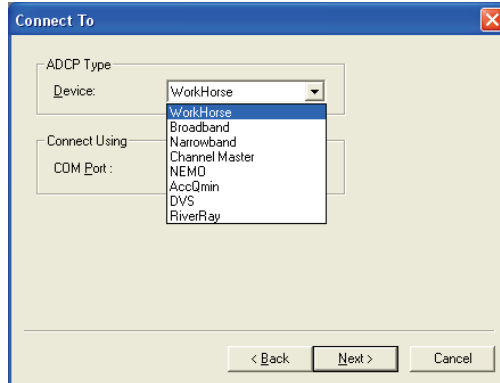


Figure 1. BBTalk Connect To Screen

- c. On the **Port Settings** screen, select the baud rate, parity, stop bits, and flow control. Click **Next**.

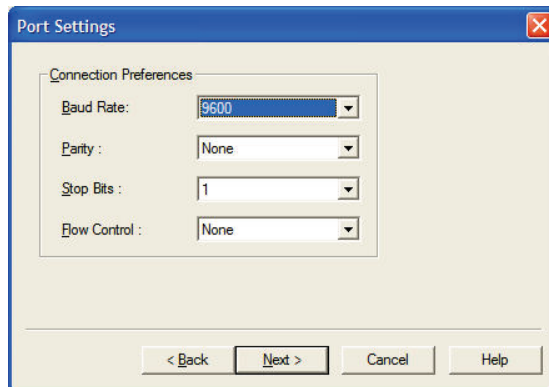


Figure 2. BBTalk Port Settings Screen



NOTE. To return to the **Port Settings Screen** while running *BBTalk*, click **File, Properties**.

- d. On the **Options** screen, select the desired settings.
 - **Send Break On New Connection** – As soon as *BBTalk* starts, it will send the ADCP a Break.

- **Use Software Break (==) with Radio Modems** – When *BBTalk* is instructed to send a Break, the "==" is sent instead.
- **Connect to Last Open Port On startup** – *BBTalk* will immediately start without going through the configuration screens. This box must be checked to enable double-clicking on a script file and have *BBTalk* automatically start and run the script.
- **Overwrite Log Files When Opening** – Log files with the same name will be overwritten.
- **Error Checking for Script Files** – Check the Script file for errors before running.
- **Send CK On Baud Rate Change (CB Command)** – Automatically sends a CK command to WorkHorse and Broad-Band ADCPs to save the baud rate after the baud rate has been changed using the CB command.
- **Echo Characters** – *BBTalk* will echo typed characters to the screen. This is required for NEMO communications, otherwise leave unchecked.
- **Wait for Prompt in Script File** – *BBTalk* will wait for the ">" prompt while running script files.

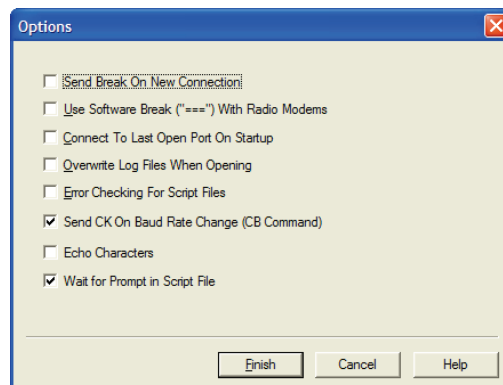


Figure 3. BBTalk Options Screen



NOTE. To return to this screen while running *BBTalk*, click **Tools, Options**.

e. Click **Finish**.

- f. On the **File** menu, click **Break** (you can also press the **End** key to send a break or use the Toolbar and press the **B** button). You should see the wakeup message appear on the log file window.

```
[BREAK Wakeup A]
xxxxxx ADCP Version xx.xx
Teledyne RD Instruments (c) 1996-2007
All Rights Reserved.
>
```

If the wakeup message is not readable, do the following.

- a. On the **File** menu, click **Properties**.
- b. Click the **Auto Detect ADCP** button.
- c. Click **OK** when the ADCP is detected. Try to wakeup the WorkHorse again.

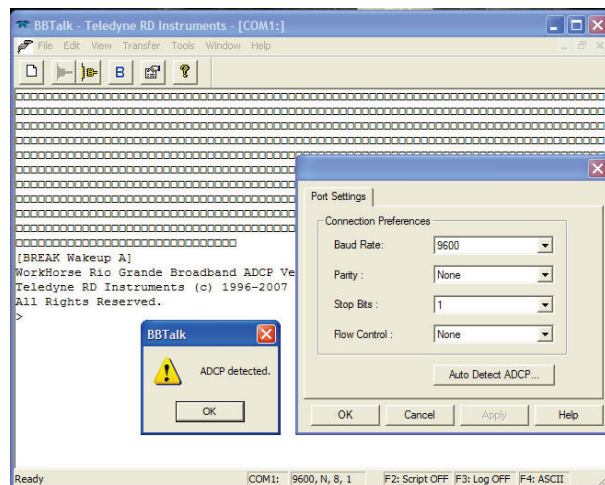


Figure 4. BBTalk Auto Detect

2.3 Sending Commands to the ADCP

- a. Setup the communication parameters between *BBTalk* and the ADCP.
- b. Wake-up the ADCP by pressing **End**.
- c. At the ">" prompt in the communication window, enter the direct command you wish to send to the ADCP and then press the **Enter** key. Refer to the Command and Output Data Format Guide for a listing of all direct commands and their format.

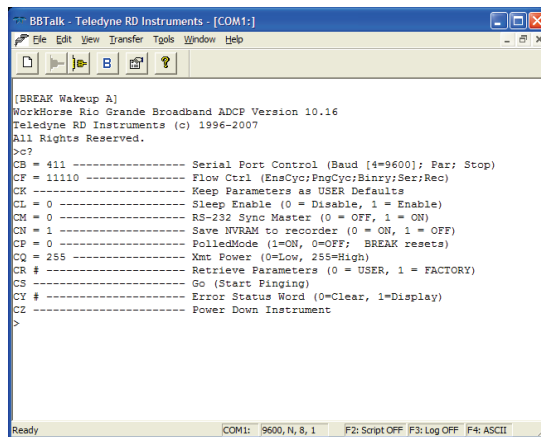


Figure 5. Sending a Command to the ADCP Using *BBTalk*

You may also send commands or verify if a command was sent using the **Command History** function.

- On the **Transfer** menu, click **Command History**.
- Use the scroll bar to view all of the commands sent to the ADCP.
- To *resend a command*, select a command in the list and click **OK**.
- To *send a new command*, type the command in the window and click **OK**.

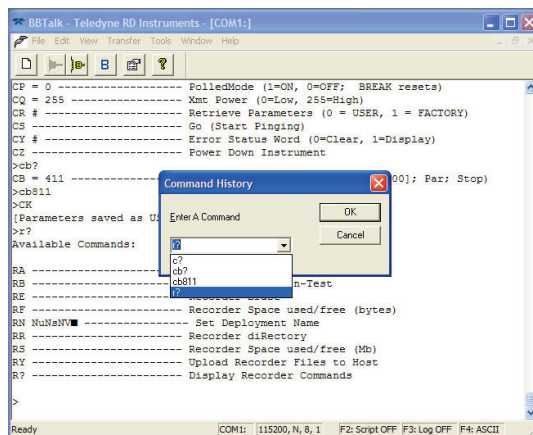


Figure 6. BBTalk Command History Screen

2.4 Using Direct Commands to Deploy your ADCP

TRDI recommends that you use our software programs *WinSC*, *VmDas*, or *WinRiver II*, etc. as your primary method of deployment. If this is not possible in your deployment then we ***strongly recommend*** that the commands shown in Table 2 be the ***minimum*** commands you send to the instrument.



CAUTION. TRDI does not recommend the use of direct commands as your primary way of deploying ADCPs as **any incorrect command setting can have severe consequences to your data collection.**

See the WorkHorse Commands and Output Data Format Guide for information on each command setting.

Table 2: ADCP Minimum Required Commands for Deployments

Command	Description
CR1	This command will set your ADCP to a known factory default setting and must be your first command
Special WM commands here – after CR1 command and before any other commands	
CFxxxx	This command will set your ADCP collection mode; binary, recorder, etc.
EAXxxx	This command will set your magnetic compass offset for true north
EDxxx	This command will set your ADCP depth
ESxx	This command will set your ADCP's expected salinity
EXxxxx	This command will set your ADCP's coordinate system; earth, beam, etc.
EZxxxxxx	This command will set what sensors will be used by your ADCP; heading, pitch, roll, temp, etc.
WBx	This command will set the water profile bandwidth between wide (0) and narrow (1)
WNxx	This command will set the number of depth cells to collect
WPxx	This command will set the number of pings to average
WSxxxx	This command will set the depth cell size to use
TExxxxxxx	This command will set the time between ensembles
TPxxxxxx	This command will set the time between pings
CK	This command will save your setup to the internal RAM and must be your second to last command
CS	This command will start your deployment and must be your last command



CAUTION. Although these are our recommended minimum commands, **they may not be the only commands you need for your deployment to be successful!**



CAUTION. Read the warning on sending the WM 1 or WM15 commands in the WorkHorse Commands and Output Data Format Guide (for firmware versions 16.30 and higher).



NOTE. Your deployment may require additional commands and these commands can be sent after the CR1 and any special WM commands but must be placed before the CK command.



CAUTION. Disconnect a pinging ADCP from the COM port before shutting down your computer. The computer might send an unintended break to the instrument.



CAUTION. Do not reconnect an ADCP that is pinging to the COM port until after the computer is booted and ready. Otherwise, the pinging ADCP will be detected as a mouse and can cause the cursor to move all over the screen (also called “crazy mouse”).

Other than rebooting the computer, there is no way to stop the ADCP from being detected as a mouse once it has started.

2.5 Recovering Data from the ADCP's Recorder

- a. Send a BREAK to wake up the ADCP, and then send CR1.
- b. On the **File** menu, click **Recover Recorder**.



NOTE. This option is available for WorkHorse and BroadBand ADCPs only.

- c. Enter the directory where the files will be downloaded. Click **OK**.
- d. Select the file to recover by clicking on the file name or check the **Select All Files** box to recover all of the files. Click **OK**.

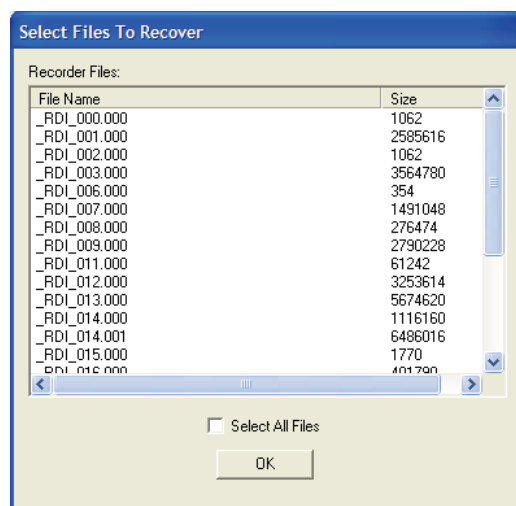


Figure 7. Select Files to Recover Screen

- e. *BBTalk* displays current protocol status, filename being received, total size of receiving file and current number of bytes received. To speed up data recovery, check the **Disable Window Output** box.

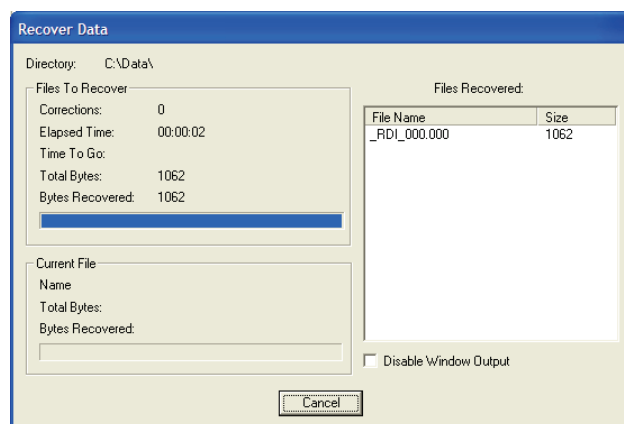


Figure 8. Recovering Data from the ADCP's Recorder

2.6 Recover Data Using Direct Commands

Another option is to use direct commands to recover data from the Work-Horse recorder. The following command sequence should be used.

- a. Send a BREAK to wake up the ADCP, and then send CR1.
- b. Send a RR command. The ADCP will display the recorder's contents.
- c. Send a RY## command (where ## is the file number you wish to recover). The RY command will start the Y-Modem protocol routine. Find your file in the recorder contents display. Ignoring the volume name, FORMAT, and DIR listings at the top of the directory, as well as data files containing 0 bytes, count down from the top (first non-zero) data file to your file. If you see that your file is the fifth file in the recorder, use RY05.

Once you send the RY## command you will be prompted to select your recording directory path to save the recorder data. When you click OK, the download will begin. When your file is downloaded, the *BBTalk* program will display "Recovery Complete".



NOTE. Use RY (without the file number) to recover the entire recorder data.

- d. Repeat step "c" to recover another data file.

2.7 Recovering Data from the Loop Recorder

- a. On the **File** menu, click **Recover Loop Recorder**.



NOTE. This option is available for AccQmin (SWADFM/DWADFM and VADCP), Channel Master, and RiverRay ADCPs only.

- b. Enter the directory where the files will be downloaded. Click **OK**.
- c. *BBTalk* displays current protocol status, filename being received, total size of receiving file and current number of bytes received.

2.8 BBTalk Script Files

You can control what *BBTalk* sends and captures by writing *BBTalk* script files. Script files are simply ASCII files produced by ASCII editors such as NotePad® (see [Example Script File](#)). In general, they contain ASCII characters that are sent out through the serial port. In addition, *BBTalk* script files can contain embedded commands that control the behavior of *BBTalk* as it interprets the script file.

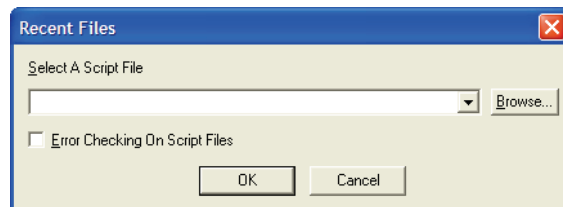
- All lines in the script file except those beginning with a '\$' or a ';' character are sent out through the serial port delimited with a Carriage Return <CR> (A line feed character <LF> = ASCII 10 decimal is NOT sent).
- Lines containing a semi-colon ';' = ASCII 59. If the first character of a line containing a semi-colon is not a '\$' character, then all characters preceding the semi-colon are sent followed by a <CR>. All characters following the semi-colon (including the semi-colon) are ignored. This feature is to provide file comments that the user may insert for script file clarity.
- Lines beginning with a dollar sign '\$' = ASCII 36 decimal are script file control commands (see Table 3).

2.8.1 Running Script Files

To run a script file, press <F2>. Select the script file to run from the scroll-down list. If no extension is given for the script file, an extension of *.txt is assumed.



NOTE. Script files can have *.rds, *.txt, *.scr, or any other extension as long as they are ASCII text files. Double-clicking a *.rds file will automatically start *BBTalk* and run the script file if the **Connect to Last Open Port** is selected on the **Options** screen (Figure 3).



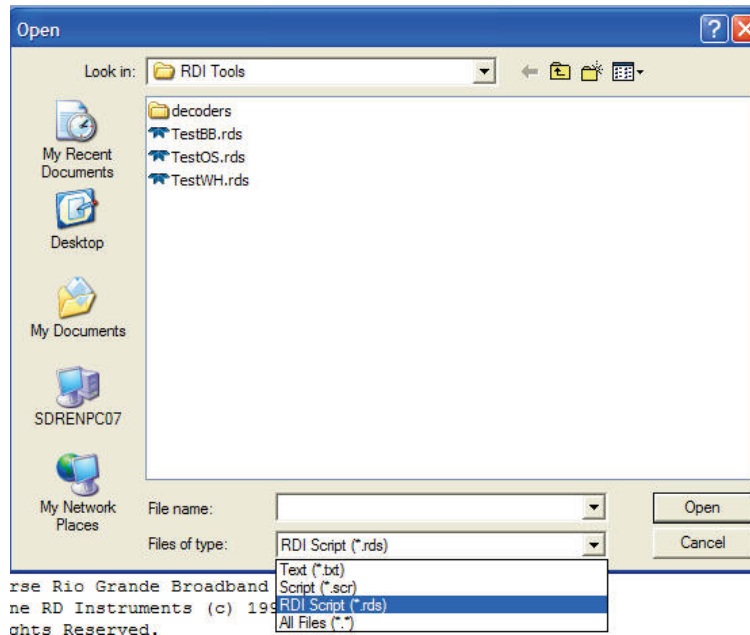


Figure 9. Selecting a Script File to Run



NOTE. When a script file is running, the status bar will change to **F2: Script On.**

2.8.2 Writing Script Files

To write your own script file, do the following.

- a. Create an ASCII text file. Use a text editor such as NotePad®. Use the *.rds extension when saving the file.



NOTE. Double-clicking a *.rds file will automatically start *BBTalk* and run the script file if the **Connect to Last Open Port** is selected on the **Options** screen (Figure 3).

- b. To send a command to the ADCP, use one command per line. *BBTalk* will automatically add a carriage return after the command is sent.
- c. To add comments to your script file, add a semicolon to the beginning of the line. *BBTalk* will ignore all comments.
- d. To use a *BBTalk* script file command (see Table 3), add one command per line.

2.8.3 Script Commands

The following are legal script commands:

Table 3: BBTalk Script Commands

Command	Description
\$A	Switch to ASCII display.
\$B	Sends a <BREAK>.
\$COM:bbbb,p,d,s	Sets up BBTALK communication parameters: bbbb = baud rate (1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200) p = parity (N = none, E = even, O = odd) d = number of data bits (7 or 8) s = number of stop bits (1 or 2)
\$Dnnnn	Delays nnnn seconds. For example, \$D0.5 will delay 0.5 seconds.
\$F	Finds the correct COM Port settings of the ADCP.
\$H	Switch to HEX display.
\$Lfilename.ext	Toggle logging. If the file name is missing and logging is to be turned on, then a window pops up asking for the log file name. Type in the file name and press the enter key to continue. If the file already exists, data will be overwritten.
\$lfilename.ext	Same as above, except a lower case "L" is used. If the file already exists, data will be appended.
\$Ptext to screen	Prints "[text to screen]" on the screen and also in log file if logging (useful for comments).
\$N	Allows you to send Narrowband binary command syntax.
\$R	Repeats script file (remainder of script file is ignored).
\$Sxx	Send binary data (in HEX format). For example, \$S41 will send "A" to the ADCP.
\$T	Sets the ADCP clock to the PC clock time. The \$T command needs a blank line proceeding it.
\$tcccc	Sends any text after the "\$" and a CR/LF. For example, \$t\$GGA will send \$GGA and a CR/LF to the ADCP.
\$Wddd,nnn	Wait for character ddd ASCII (Decimal representation). If nnn is missing, it waits forever. \$W62 means wait for the ">" prompt forever.
\$W"c",nnn	The same as above, but "c" is represented as a character. \$W">",2 means wait for the prompt for a maximum of two seconds and then continue.
\$X	Exits BBTalk.

2.8.4 Example Script File

The following example is a printout of the script file TestWH.rds (see [Using a Script File to Test the ADCP](#)).

```

;-----
;                                TESTWH.RDS
;      Script file for testing RD Instruments WorkHorse
;      ADCP with the BBTalk program
;      Copyright (C) 2001 by RD Instruments - All rights reserved
;      Licensed for exclusive use with RD Instruments products or data
;-----
;-----

; TRDI - WH ADCP testing script file:
; FILE name = "TESTWH.RDS"
; 1 November 2001
; Modified by TRDI on:
; none
$LWH_TESTS.TXT
$P =====
; Print the following lines.
$P WH ADCP Test
$P *****
$P

```

```

$P The following tests are basic tests which will confirm that your system
$P is ready for use. Some tests will need to be run with the system in
$P water. You will be prompted when this is necessary.
$P
$P Connect the WH ADCP to power and the PC as described in the manual.
$P Turn on power to the WH ADCP.
$P
$P The results of all tests will be printed to the screen and saved to the
$P log file WH_TESTS.TXT. A file called WH_TESTS.TXT with the results of
$P this test will be created in the same directory as the BBTalk program
$P is running from.
$P
$P The following tests will be performed:
$P
$P PA Basic Internal System Tests
$P PC2 Sensor Verification Test
$P PC1 Beam Continuity Test
$P
$P Program is delaying 20 seconds before continuing.
$P Press <Enter> to continue.
$W62,20
$P =====
$P
$P PA -- Basic Internal System Tests
$P The following tests will verify that the internal electronics are
$P performing correctly. These tests are best run when the transducer
$P face is submerged in water. A bucket of water deep enough to cover
$P the transducer beams is all that is needed. If done in air some tests
$P may fail.
$P
$P Program is delaying 10 seconds before continuing.
$P Press <Enter> to continue.
$W62,20
$P =====
$P
$P Sending a break to Wake Up the System
$B
$D2
$P =====
$P
$P Restoring factory defaults into temporary memory for TEST.
$P
$P CR1
$D1
$P =====
$P
$P Collecting system specific data.
$P
$P TS?
$D1
$P0
$D1
$P3
$D1
$P =====
$P
$P Starting the Automated Tests.
$P
$W62,100
$P
$P All of the above tests should have passed. Review the file WH_TESTS.TXT
$P to verify your tests results. Remember that some tests will fail
$P unless the transducer is immersed in water. Consult your Technical
$P Manual for trouble shooting tips if this test did not pass.
$P
$P
$P Program is delaying 15 seconds before continuing.
$P Press <Enter> to continue.
$W62,15
$P =====
$P
$P PC2 -- Sensor Verification Test
$P The following test will confirm that your heading, pitch, roll,
$P temperature, orientation sensor, and pressure sensor (if installed) are
$P operating. You should turn and tilt the ADCP and confirm that changes
$P occur in the heading, pitch, and roll. Verify that the Up/Down setting
$P agrees with the direction of your ADCP transducer. Verify that the
$P ambient temperature and pressure are reasonable values.

```

```

$P
$P This test in itself does not calibrate or confirm the accuracy of the
$P sensors. However, if you turn and till the ADCP while comparing the
$P output to a known reference then you can confirm the accuracy.
$P
$P The following test is best run when the transducer face is in air and
$P the transducer is pointing the direction you intend to deploy the
$P instrument.
$P
$P This test will continue to run until you stop the test.
$P
$P Program is delaying 20 seconds before continuing.
$P Press <Enter> to continue.
$W62,20
$P =====
$P
$P Sending a break to Wake Up the System
$B
$D2
$P =====
$P
PC2
$W62,1000
$P
$P The Sensor test is complete. The heading, pitch, roll sensors should
$P have changed as you turned and tilted the system. The Up/Down setting
$P should have agreed with the direction of your ADCP transducer. The
$P ambient temperature and pressure (if installed) should have been
$P reasonable values.
$P
$P All of the above tests should have passed. Review the file WH_TESTS.TXT
$P to verify your tests results. Consult your Technical Manual for trouble
$P shooting tips if this test did not pass.
$P
$P Program is delaying 10 seconds before continuing.
$P Press <Enter> to continue.
$W62,10
$P
$P =====
$P
$P PC1 -- Beam Continuity Test
$P The following test will confirm that each of the beams on your
$P transducer is capable of receiving signals. This test must be run in
$P air and free of external interference to pass.
$P
$P This test will require you to rub each of the beams on the transducer.
$P This is done with quick rubbing movements across each of the urethane
$P faces.
$P
$P Program is delaying 25 seconds before continuing.
$P Press <Enter> to continue.
$W62,25
$P =====
$P
$P Sending a break to Wake Up the System
$B
$D2
$P =====
$P
PC1
$W62,1000
$P
$P The Beam Continuity test is complete. Each of the beams should have
$P passed. Review the file WH_TESTS.TXT to verify your tests results.
$P Consult your Technical Manual for trouble-shooting tips if this test
$P did not pass.
$P
$P =====
$P All tests have been run and if passed your system is ready for
$P deployment.
$L

```

2.8.5 Using a Script File to Test the ADCP

- Connect the ADCP to the computer as described in the appropriate ADCP User's Guide.
- Setup communication parameters between *BBTalk* and the ADCP.
- Click **File**, **Send a Break** to send the wakeup command (BREAK) to the ADCP
- On the **File** menu, click **Send Script File**. Click the **Browse** button "...".
- Select the appropriate script file (see Table 4).

Table 4: BBTalk Test Script Files

Script File Name	ADCP Type	Results Saved to
TestBB.rds	Broadband	BB_TESTS.txt
TestOS.rds	Ocean Surveyor	OS_TESTS.txt
TestWH.rds	Workhorse	WH_TESTS.txt



NOTE. These script files (*.rds) were copied into the same directory as *BBTalk* when you installed the RDI Tools software CD sent with your system.

Double-clicking a *.rds file will automatically start BBTalk and run the script file if the **Connect to Last Open Port** is selected on the **Options** screen (Figure 3).

- Follow the prompts on the screen (see Figure 10).
- To review the test results, open the results log file (*.txt) with any text editor (i.e. Notepad).

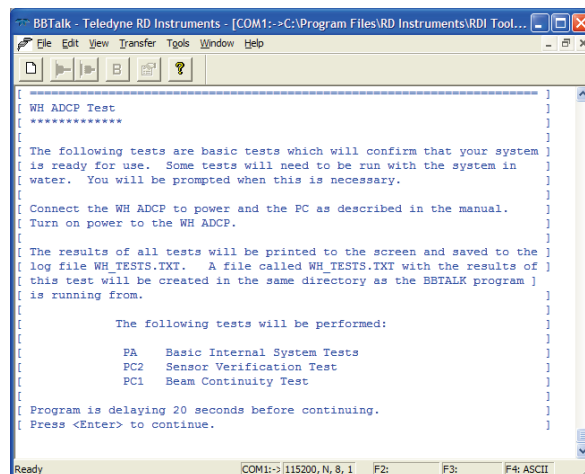


Figure 10. Running a Script File to Test the ADCP

The *BBTalk* program saves the test results file to different locations based on how the program was started. When you start *BBTalk* from the desktop icon and run the test script file, the result log file is created on the desktop. If you run *BBTalk* from the start menu, the results file is put in C:\Documents and Settings\All Users\Start Menu\Programs\RD Instruments\RDI Tools. It is only when you double-click the *.rds file in the RDI Tools folder that the results are saved to the RDI Tools folder.

To make sure the result file is always saved to the same location, do the following.

- a. Start *BBTalk*.
- b. On the **File** menu, click **Edit Script File**.
- c. Select the *TestBB.rds*, *TestOS.rds*, or *TestWH.rds* file (the example below shows the *TestWH.rds* file being edited).
- d. Add a directory path in front of the results file name. For example, edit the \$L line from \$LWH_TESTS.TXT to \$LC:\Program Files\RD Instruments\RDI Tools\WH_TESTS.TXT.
- e. Save the file and exit *NotePad*.

```
TESTWH.RDS
;           Script file for testing RD Instruments WorkHorse
;           ADCP with the BBTalk program
;           Copyright (C) 2001 by RD Instruments - All rights reserved
;           Licensed for exclusive use with RD Instruments products or data
;-----
; TRDI - WH ADCP testing script file:
; FILE name = "TESTWH.RDS"
; 1 November 2001
; Modified by TRDI on: 10 April 2006
; Added path to result file
$L C:\Program Files\RD Instruments\RDI Tools\WH_TESTS.TXT
$P =====
↓
```

2.9 BBTalk LOG Files

The LOG feature lets you record data to a disk file. You name the file by pressing the <F3> key. You can enable logging at any time. The status bar shows the Log status. To use Log, do the following steps.

- a. Press <F3> to enable the LOG function.
- b. Select the Log File from the scroll-down list or use the **Browse** button to create a new log file.
- c. If the file already exists, the program asks you if you want to overwrite the existing file unless the **OverWrite Log File** box is selected.
- d. All data sent to the screen will now be written to the file you specified. You can enable the Log feature at any time, even if the ADCP is already sending data.

- e. To disable Log, press <F3>.

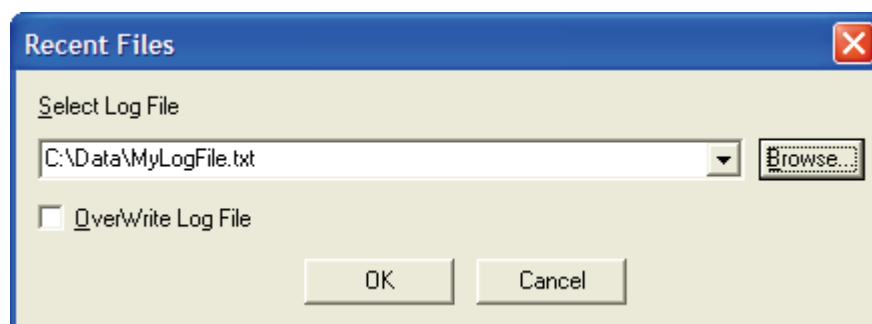


Figure 11. Log Files

3 Using BBLIST

BBLIST takes the binary data files created by the ADCP and lets you convert selected data fields to common units of measurement in an ASCII-text format. You can then use the ASCII files in programs that accept this format. *BBLIST* never alters your original ADCP binary data files. *BBLIST* lets you:

- Set processing parameters.
- Transform velocity data.
- Select the display/conversion limitations (e.g., data fields bin range, ensemble range, output file size).
- Select the ADCP data fields to display or convert.
- Monitor and control the conversion process.
- Look at radial beam data.

3.1 Starting BBLIST

To start *BBLIST*, type the following at the DOS prompt.

- **BBLIST** - Starts *BBLIST* and lets you select the ADCP binary data file from within the program.
- **BBLIST C:\BBDATA\TEST.000 /M** - Starts *BBLIST* and loads the *TEST.000* file located on the C: drive in the BBDATA directory. The /M forces *BBLIST* to use monochrome screens.

When you run the program the first time, *BBLIST* displays its introduction and copyright screen. During operation, *BBLIST* creates/updates a pointer file named *BBLIST.PTR*. This file saves the current working directory, the name of the last binary file used, the name of the format file (*.FMT) last used, and the color selection. This may help save you time by automatically calling up the files you used last. If you need to see the introduction screen again, you must first delete or rename the *BBLIST.PTR* file.

Extensive help is available while using *BBLIST* by pressing <F1>. The help screen lists all of the menus and a description of each option available. You may also print the help screen file by printing the file *BBLIST.DOC*.

3.2 BBLIST Menus

BBLIST has four main menus (Figure 12) to guide you through the steps needed to convert a ADCP binary raw data file to an ASCII data file.

File Menu. Use this menu to save your menu settings and conversion layout selections to a configuration format file (*.FMT), to load a ADCP binary data file or format file, to display the file information screen, or to exit *BBLIST*.

Process Menu. Use this menu to set the processing parameters for *BBLIST*. You can set the velocity reference, magnetic variation (declination), velocity measurement units, and depth measurement units.

Display Menu. Use this menu to select the type of data to view. Data available for display include ADCP setup, sensors, reference layer, bottom-track, and profile.

Convert Menu. Use this menu to convert the binary ADCP data set into ASCII. Before starting the conversion process, you can define the conversion limitations and the output format.

File Process Display Convert				BBLIST 3.11	
Use <↑><↓> to move to the file, then press <Enter> to load					
1 BINARY FILE					
Name	Ext	Size			
EXAMPLE	000	245,482			
			ENSEMBLE INFO IN DATA SET		
			Number	Date	Time
			First 1	28-SEP-95	14:50:00
			Last 573	02-OCT-95	14:10:00
			ENSEMBLE INFO IN EXAMPLE.000		
			Number	Date	Time
			First 1	28-SEP-95	14:50:00
			RECORDED DATA		
			Leader		
			Velocity (earth)		
			Correlation		
			Intensity		
			Percent Good		
FORMAT FILE NAME			Ensemble size 414 bytes		
F1-Help <↵>-Reload			F10-Menu		

Figure 12. BBLIST Display

3.3 Using BBLIST to Convert Files

This is an example of how to convert the binary file *EXAMPLE.000* to ASCII delimited text. For this example, we have chosen to extract only the magnitude, direction, and range data. Remember that *BBLIST* never changes your original data file — you can convert the raw data to as many different ASCII formats as desired.

- a. Start BBLIST. To start *BBLIST*, type BBLIST at the DOS prompt.
- b. Load *EXAMPLE.000* binary file. Press **F3** to load the binary file.
- c. Set the processing parameters. Use the **Process** menu (Figure 13) to set the velocity reference, magnetic variation, velocity measurement units, and depth measurement units. Use the **Space** key to toggle each setting. Make sure that **Mark Below Bottom** is set to **No**.



NOTE. BBLIST uses the direct ADCP commands for presenting depths (ED-command).

- d. Set the conversion parameters. Use the **Convert, Limits** menu (Figure 14) to set the conversion limitations and parameters. You may want to increase the Max file size to 1000 kB if you plan to back up the ASCII data to floppy disks.
- e. Define the format. Consider the following:
 - What data do you want to extract? For this example, we choose *magnitude, direction, and range*.
 - To what program will you export the ASCII data? Does the program support space, tab, or comma-delimited text? We choose space-delimited text for the example. To change, select Field delimiter and press the **Space** key to toggle between **Tab**, **Comma**, and **Space**.
 - Do you want vertical or horizontal placed data? For this example, we choose a vertical data format. To change, select Bin layout and press the **Space** key to toggle between **Vertical** and **Horizontal**.

Begin defining the format by doing the following.

1. Select **Convert, Define Format**. When you first enter this screen, *BBLIST* displays only a **Format End** marker. To begin entering data fields, press **<End>** to display the **Define Format** selection menu (Figure 15). If you make a mistake, select delete block and re-enter your choice.
2. Select **Ensemble Info** and press **<end>**. Select **Number** and press **<End>**.
3. Move the cursor to the **Format End** marker and press **<End>**.

4. Select **Profiles, Mag and Dir**. Select **M** for magnitude and press <End>.
 5. Move the cursor to **Line End** and press <End>.
 6. Select **Profiles, Mag and Dir**. Select **D** for direction and press <End>.
 7. Move the cursor to **Line End** and press <End>.
 8. Select **Profiles, Depth Ref**. Select **R** for range and press <End>. Your display should look like Figure 16.
- f. View the format. Press **F9** to view the format before converting (Figure 17). Use the + and - keys to increase the ensemble number. Press **F9** again to return to the define format screen.
 - g. Save the format. For future use, save the format to a *.FMT file by pressing **F2** and naming the file (example; MYFORMAT).
 - h. Begin the conversion. Select **Convert, Start Conversion**. Enter a file name for the ASCII file data set. *Be sure to use a file name that differs from any existing file name.* Do not use the name of the binary data set. If *BBLIST* detects that the file name already exists, you are given the option to overwrite the existing files.

3.4 Using a BBLIST Format File to Convert Files

Once you have created a format file, you can use this file to convert other data files.

- a. Start *BBLIST*. To start *BBLIST*, type BBLIST.
- b. Load *EXAMPLE.000* binary file. Press **F3** to load the binary file.
- c. Load Format file. Press **F3** to load the format file.
- d. Begin the conversion. Select **Convert, Start Conversion**. Enter a file name for the ASCII file data set. *Be sure to use a file name that differs from any existing file name.* Do not use the name of the binary data set. If *BBLIST* detects that the file name already exists, you are given the option to overwrite the existing files.

File Process Display Convert BBLIST 3.11
 Use <Spacebar> to choose velocity reference: ADCP, bottom-track, ref. layer

Name	velocity Reference	ADCP	ENSEMBLE INFO IN DATA SET
EXAM	Magnetic variation	0.00	Number 1 Date 28-SEP-95 14:50:00
	Velocity units	ADCP	Last 573 02-OCT-95 14:10:00
	Depth units	ADCP	
	mark below Bottom	NO	

ENSEMBLE INFO IN EXAMPLE.000

First	Number	Date	Time
1	1	28-SEP-95	14:50:00

RECORDED DATA

Leader
 Velocity (earth)
 Correlation
 Intensity
 Percent Good

Ensemble size 414 bytes

FORMAT FILE NAME

F1-Help F10-Screen

Figure 13. Set the Processing Parameters

File Process Display Convert BBLIST 3.11
 Select the bin numbers to convert to ASCII

1 BINARY FILE

Name	Ext	S	Limits	ENSEMBLE INFO IN DATA SET
EXAMPLE	000	245,	Bin selection 128-1 (skip 0)	Number 1 Date 50:00 10:00
			Add data to missing bins	NO
			Ensemble increment	1
			Pruning method	ENSEMBLE
			Ensembles 1-65535	Time 50:00
			Write missing data from ADCP	NO
			Go auto to next file in set	YES
			Max file size [kB]	1000

Velocity (earth)
 Correlation
 Intensity
 Percent Good

Ensemble size 414 bytes

FORMAT FILE NAME

F1-Help F10-Screen

Figure 14. Set the Conversion Limitations and Parameters

File Process Display Convert BBLIST 3.11
 Insert ensemble number or ensemble recording time information

FORMAT FILE -

1	2	3	4	5	6	7
FORMAT END						

Bin layout: Vertical

Insert:

Ensemble info
 Sensors
 Reference layer
 bottom Track
 Profiles
 User comment
 insert Line

Clear table
 Delete block

Field delimiter Space

Line: no. - total len. - field: no. - len. -
 Data from EXAMPLE.000 : format , , , , , , ,

F1-Help <-Change F9-View F10-Menu

Figure 15. Define Format Selection Menu

File Process Display Convert				BBLIST 3.11			
Use <↑><↓><+><-> to move bar, <Enter> to change fields, <F9> to view format							
FORMAT FILE							
1	2		3	4	5	6	7
1	Ens no.	LINE END					
2	Mag 15	Dir 15	Range 15	LINE END			
3	Mag 14	Dir 14	Range 14	LINE END			
4	Mag 13	Dir 13	Range 13	LINE END			
5	Mag 12	Dir 12	Range 12	LINE END			
6	Mag 11	Dir 11	Range 11	LINE END			
7	Mag 10	Dir 10	Range 10	LINE END			
8	Mag 9	Dir 9	Range 9	LINE END			
9	Mag 8	Dir 8	Range 8	LINE END			
10	Mag 7	Dir 7	Range 7	LINE END			
11	Mag 6	Dir 6	Range 6	LINE END			
12	Mag 5	Dir 5	Range 5	LINE END			
13	Mag 4	Dir 4	Range 4	LINE END			
14	Mag 3	Dir 3	Range 3	LINE END			
Line: no. 2 total length 25				Field: no. 4 len. 2			
'End of line: <CR><LF>'				Format ''			
Data from EXAMPLE.000 :							
79 216.7 3221							
.....1.....2.....3.....4.....5.....6.....7.....8							
F1-Help <-Change				F9-View F10-Menu			

Figure 16. Defining the Format

File Process Display Convert				BBLIST 3.11			
Use cursor control keys to move bar, <Tab>/<Shift-Tab> to scroll left/right							
ENSEMBLE 38				TIME 28-SEP-95 21:00:00.00 BINARY FILE EXAMPLE.000			
38							
154	175.5	3221					
161	173.6	3021					
160	178.2	2821					
184	177.5	2621					
208	174.8	2421					
213	179.2	2221					
91	181.3	2021					
122	185.2	1821					
228	183.8	1621					
203	184.8	1421					
186	182.2	1221					
194	175.6	1021					
193	172.8	821					
195	175.6	621					
184	174.1	421					
F1-Help -/+-Ensemble J-Jump A.T.L.V...-Display				F9-Format F10-Menu			

Figure 17. View the Format before Converting

3.5 Report File

When the conversion process is complete, *BBLIST* creates an ASCII report file (*.RPT). You can view this file with any text editor. This file contains the following information about the settings and data in the converted files.

- **ADCP information** - system frequency, beam angle, number of profiling beams, transducer orientation (up/down), transducer pattern (concave/convex), transducer connection (connected/disconnected), and CPU firmware version number.
- **ADCP setup** - number of bins, bin length, blank after transmit length, pings per ensemble, time per ping, and profiling mode.
- **ASCII file data format** - a description on the contents of each line in the converted data file (for one ensemble).
- **Processing parameters** - velocity units, velocity reference, depth units, bin sequence, magnetic variation.

Sample BBLIST report file:

REPORT FOR ASCII DATA CONVERSION

1. ADCP INFORMATION:

Frequency 300 kHz
Beam angle 20 deg
4 beam system
Up-looking orientation
Convex beam pattern
Transducer head connected
CPU firmware 8.01

2. ADCP SETUP:

Number of bins 15
Bin length 200 cm
Blank after transmit 200 cm
Distance to first bin 421 cm
Transmit length 207 cm
Pings per ensemble 300
Time per ping 1.99 s
Profiling mode 1

3. ASCII FILE DATA FORMAT:

Line 1: Ensemble number
Line 2-16: Magnitude, Direction, Bin range

4. PROCESSING PARAMETERS:

Velocity units: ADCP
Velocity reference: BT
Depth units: ADCP
Bins: From 128 to 1 skip 0 bin
Magnetic variation 0.00 deg
Do not mark data below bottom

END OF REPORT

3.6 BBBATCH Program

When you have mastered the *BBLIST* program, you can use *BBBATCH* to convert binary data sets to ASCII data sets in a DOS batch mode. This comes in handy when you have several data sets to convert or have a large data set that you want to convert overnight. Here is the syntax for *BBBATCH*:

BBBATCH BinaryFileName FormatFileName AsciiFileName

BinaryFileName = name of the binary data set to convert (no extension needed)

FormatFileName = name of the format file to use

AsciiFileName = name of the ASCII data set (unique name)

Running *BBBATCH* without any command line parameters displays the syntax.

4 Utility Software

The following Windows® programs have been provided to supplement data processing. These programs have been installed to the directory C:\Program Files\Rd Instruments\RDI Tools.

4.1 Using BBSub

BBSub is a raw data file sub-sectioning utility. This allows you to save a portion of a large raw data file as a separate raw data file. *BBSub* starts copying ensembles from the input file to the output file starting with the ensemble specified by the **Start Ensemble Number**. If the **Start Ensemble Number** is specified to zero, *BBSub* starts copying from the first read ensemble. *BBSub* will continue copying ensembles until it reaches ensemble number **End Ensemble Number**. If the **End Ensemble Number** is specified to zero, it copies until the end of the file. Only ensembles with valid checksum are copied.

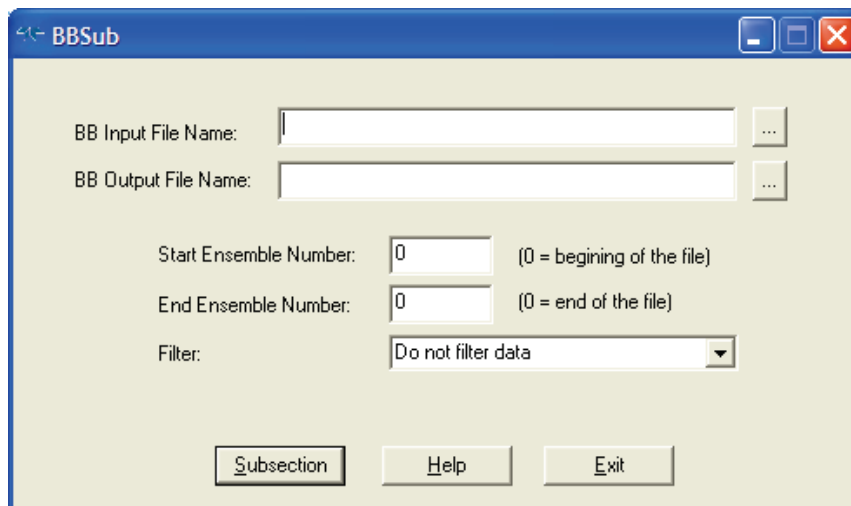


Figure 18. BBSub Display

To subsection a file, do the following.

- Enter the name and path of the BroadBand ADCP raw data in the **BB Input File Name** box or use the browse button “...” to locate the file. *BBSub* “remembers” the input and output directory if the **Browse** button “...” is selected.
- Enter the name and path of the subsectioned file in the **BB Output File Name** box or use the **Browse** button “...” to locate the file.
- Enter the **Start Ensemble Number**.
- Enter the **End Ensemble Number**.

e. Select the method to filter the data:

- **Do not Filter Data** – No filtering is done.
- **Extract only Down-Looking Data** – Only ensembles with orientation sensor pointing down will be copied.
- **Extract only Up-Looking Data** – Only ensembles with orientation sensor pointing up will be copied.
- **Force Sensor to be Down-Looking** – All the ensembles will have the orientation sensor pointing down.
- **Extract Secondary OS Data** – If both the BroadBand and NarrowBand data are recorded (Ocean Surveyor) this option extracts the NarrowBand data. This makes the data compatible with TRDI's standard software.
- **Redo Ensemble Numbers** – Ensembles will be renumbered starting with ensemble number 1.
- **Resample Data** – Every n-th ensemble can be extracted to a new binary data file.
- **Force 3 beams to 4** – Changes the number of beams listed in fixed leader of pd0 from 3 to 4. This is for horizontal data that has 3 beams listed in the fixed leader and cannot be re-processed by other programs (see FSB-188).

f. Press the **Subsection** button to begin.

g. Press the **Exit** button to exit *BBSub*.

You can run *BBSub* in the batch mode by writing a simple DOS *.bat file that contains command lines with the files that you want to process. For each file to be processed, add a line to the batch file specifying the name of the raw data file to be subsectioned, the name of the output file, and the starting and ending ensemble numbers.



NOTE. For unattended operation, make sure that output file does not exist; otherwise the batch file will pause to confirm overwriting the file.

Sample Batch file

```
"C:\Data\BBSub.exe" -in:InputFile -out:OutputFile -start:StartEns
-end:EndEns -filter:OnlyUp
```

Syntax for filter:

```
-filter:OnlyDown
-filter:OnlyUp
-filter:ForceDown
-filter:OS
-filter:Redo
-filter:Resample,n where "n" is number of ensembles
```



NOTE. DOS does not like file names with spaces. Copy the BBSub.exe file and your data files to a folder with a very short path (for example, C:\Data).

4.2 Using BBConv

BBConv is a BroadBand raw data to ASCII conversion program. It uses a decoder file (.DEC) to determine how to convert the data (see [Using Decoder Files](#)). A decoder file contains interpreted instructions for converting to ASCII.

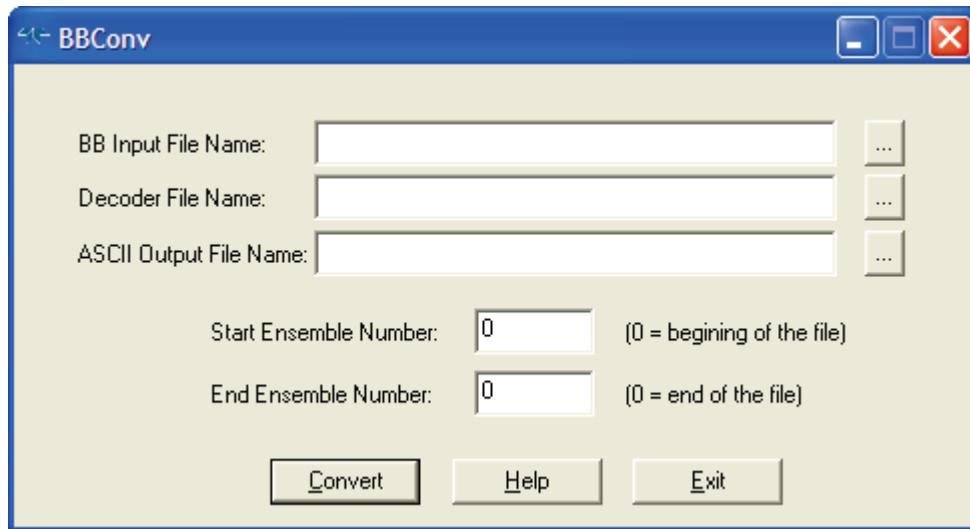


Figure 19. BBConv Display

To convert a file, do the following.

- Enter the name and path of the BroadBand ADCP raw data in the **BB Input File Name** box or use the **Browse** button “...” to locate the file. *BBConv* “remembers” the input and output directory if the **Browse** button “...” is selected.
- Enter the name and path of the decoder file in the **Decoder File Name** box or use the browse button “...” to locate the file.
- Enter the name and path of the ASCII file in the **ASCII Output File Name** box or use the **Browse** button “...” to locate the file.
- Enter the **Start Ensemble Number**.
- Enter the **End Ensemble Number**.
- Press the **Convert** button to begin.
- Press the **Exit** button to exit *BBConv*.

You can run *BBConv* in the batch mode by writing a simple DOS batch file (*.bat) that contains command lines with the files that you want to process. For each file to be processed, add a line to the batch file specifying the name of the raw data file to be converted, the name of the decoder file, the name of the output file, and the starting and ending ensemble numbers.



NOTE. For unattended operation, make sure that output file does not exist; otherwise the batch file will pause to confirm overwriting the file.

Sample Batch file

```
"C:\Data\BBConv.exe" -in:InputFile -dec:DecoderFile -out:OutputFile
```



NOTE. DOS does not like file names with spaces. Copy the *BBConv.exe* file and decoder file to a folder with a very short path (for example, C:\Data).

4.3 Using BBMerge

BBMerge merges the ASCII comma delimited format data (created by using *BBConv*), back into the raw data file format, resulting in a new raw data file.



NOTE. The ASCII text file and the BroadBand data input file should have the same number of records. If the text file contains fewer records, then zero values will be merged into the remaining BroadBand data file records. If the BroadBand data file has fewer records, then the remaining ASCII text file will be ignored.

Figure 20. *BBMerge* Display

To merge a file, do the following.

- a. Enter the name and path of the BroadBand ADCP raw data in the **BB Input File Name** box or use the **Browse** button “...” to locate the file. *BBMerge* “remembers” the input and output directory if the **Browse** button “...” is selected.
- b. Enter the name and path of the decoder file in the **Decoder File Name** box or use the **Browse** button “...” to locate the file.
- c. Enter the name and path of the ASCII file in the **ASCII File Name** box or use the **Browse** button “...” to locate the file.
- d. Enter the name and path of the new BroadBand ADCP raw data file in the **BB Output File Name** box or use the **Browse** button “...” to locate the file.
- e. Enter the **Start Ensemble Number**.
- f. Enter the **End Ensemble Number**.
- g. Press the **Merge** button to begin.
- h. Press the **Exit** button to exit *BBMerge*.

You can run *BBMerge* in the batch mode by writing a simple DOS batch file (*.bat) that contains command lines with the files that you want to process. For each file to be processed, add a line to the batch file specifying the name of the raw data file to be merged, the name of the decoder file, the name of the ASCII file, the name of the new output file, and the starting and ending ensemble numbers.



NOTE. For unattended operation, make sure that output file does not exist; otherwise the batch file will pause to confirm overwriting the file.

Sample Batch File

```
"C:\Data\BBMerge.exe" -in:InputFile -dec:DecoderFile -ascii:ASCIIFile -  
out:OutputFile
```



NOTE. DOS does not like file names with spaces. Copy the *BBMerge.exe* file and decoder file to a folder with a very short path (for example, C:\Data).

4.4 Using BBCheck

BBCheck checks your data files for integrity and quality. It does some simple error and problem checking. You should use *BBCheck* on your data files before you erase the original data files on your recorder.

BBCheck performs the following tests.

- Valid checksum
- Sequential ADCP ensemble number
- Built in test code (BIT) is zero
- Configuration byte remains constant
- All data types identified

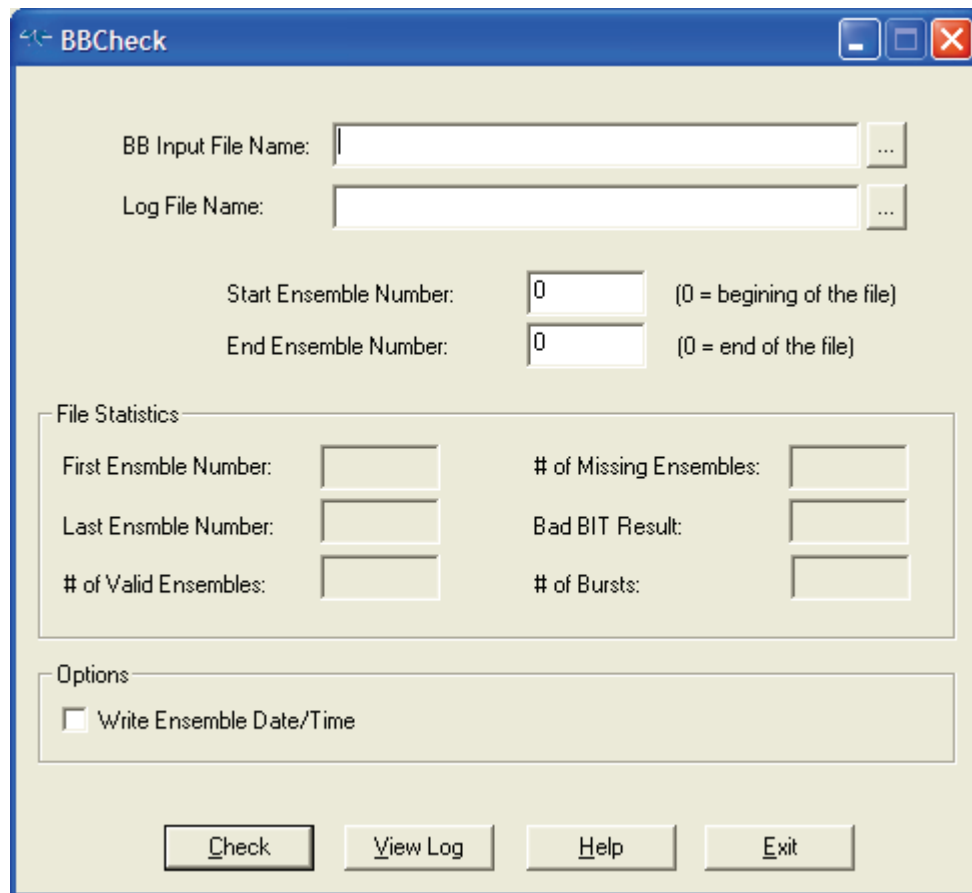


Figure 21. BBCheck Display

To check a file for errors, do the following.

- Enter the name and path of the BroadBand ADCP raw data in the **BB Input File Name** box or use the **Browse** button “...” to locate the file.

BBCheck "remembers" the input and output directory if the **Browse** button "..." is selected.

- b. Enter the name and path of the log file in the **Log File Name** box or use the **Browse** button "..." to locate the file. *BBCheck* will create the log file if it does not exist.
- c. Enter the **Start Ensemble Number**.
- d. Enter the **End Ensemble Number**.
- e. Press the **Check** button to begin.
- f. Press the **View Log** button to view detailed information about the file and the data ID numbers detected by *BBCheck*.
- g. Press the **Exit** button to exit *BBCheck*.

You can run *BBCheck* in the batch mode by writing a simple DOS *.bat file that contains command lines with the files that you want to check. For each file to be checked, add a line to the batch file specifying the name of the raw data file to be checked, the name of the output log file, and the starting and ending ensemble numbers.



NOTE. For unattended operation, make sure that the output log file does not exist; otherwise the batch file will pause to confirm overwriting the file.

Sample Batch file

```
"C:\Data\BBCheck.exe" -in:InputFile -out:OutputLogFile -start:StartEns  
-end:EndEns
```



NOTE. DOS does not like file names with spaces. Copy the *BBCheck.exe* file and data files to a folder with a very short path (for example, C:\Data).

4.5 Using BBSlice

BBSlice copies ensembles from the input file to an output file checking each ensemble number. If the ensemble number is not sequential, a new deployment file is opened, and the ensembles are copied into it. The user supplies an **Output File Prefix** and the deployment file name is created by adding "nnn.mmm", where "nnn" is deployment number starting with "000". "mmm" is a file extension that starts with "000" and increases if file reaches a maximum file size supplied by the user. If the maximum file size is 0 there is no limitation on the file size of the output file.

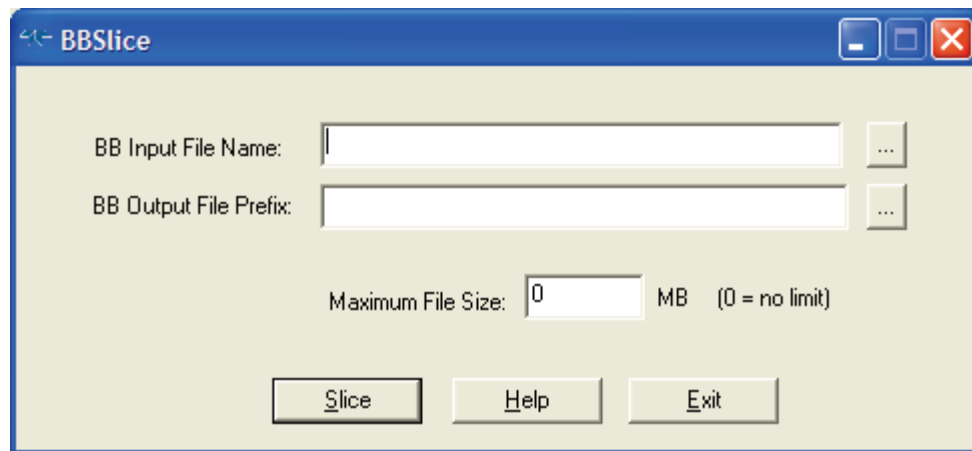


Figure 22. BBSlice Display

To slice a file, do the following.

- Enter the name and path of the BroadBand ADCP raw data in the **BB Input File Name** box or use the **Browse** button "..." to locate the file. *BBSlice* "remembers" the input and output directory if the Browse button "..." is selected.
- Enter the file name prefix in the **BB Output File Prefix** box or use the browse button "..." to locate the file.
- Enter the **Maximum File Size**.
- Press the **Slice** button to begin.
- Press the **Exit** button to exit *BBSlice*.

You can run *BBSlice* in the batch mode by writing a simple DOS *.bat file that contains command lines with the files that you want to process. For each file to be processed, add a line to the batch file specifying the name of the raw data file to be subsectioned, the name of the output file, and the starting and ending ensemble numbers.



NOTE. For unattended operation, make sure that output file does not exist; otherwise the batch file will pause to confirm overwriting the file.

Sample Batch file

```
"C:\Data\BBSlice.exe" -in:InputFile -out:OutputFile -size:MaxFileSize
```



NOTE. DOS does not like file names with spaces. Copy the BBSlice.exe file and data files to a folder with a very short path (for example, C:\Data).

4.6 Using BBSS Speed of Sound Calculator

BBSS allows you to quickly calculate the speed of sound in the water. Enter values for the water temperature, salinity, and depth. The Sound Speed and Coefficient will be calculated based on the parameters entered.

BBSS (Speed of Sound Calculator)

Expression for calculating speed of sound:

$$C = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 + (1.34 - 0.01T)(S - 35) + 0.016D$$

Parameters

Temperature (T): °C

Salinity (S): ppt

Depth (D): m

Speed of Sound

Calculated (C): m/s

Reference: (Set to C) m/s

Coefficient:

Apply Help Exit

Figure 23. BBSS Display

To calculate the speed of sound, do the following.

- Enter the water temperature in the **Temperature** box.
- Enter the water salinity in the **Salinity** box.
- Enter the water depth in the **Depth** box.
- Enter a reference speed of sound in the **Reference** box. The default is 1490 m/s.
- Click **Apply**. The calculated speed of sound and coefficient will appear in the Calculated (C) and Coefficient boxes.
- Click **Exit** to exit *BBSS*.

5 Using Decoder Files (*.DEC)

TRDI provides example DEC files that help remove user-selected data from binary data files and stores the information into ASCII comma delimited format.

A typical decoder file may contain the following:

```
; Temperature Data Decoder File
0080,3,2
0080,27,-2,0.01,2 ; Temperature (ET) in degrees
$L
```

- Lines beginning with a semicolon (;) are ignored and are for user comments.
- In addition, decoder lines may be followed by semi-colons for additional comments.
- Blank lines are ignored.
- "\$L" signifies a CR/LF should be output before outputting more data

Lines containing decoding instructions are broken down as follows:

id,offset,size,multiplier,format

where:

id	Data Structure identifier in hex (e.g. 0080). This information is presented MSB-LSB. Be careful, since the data stream outputs LSB followed by MSB.
offset	The byte # from the first byte of the data structure (e.g. The first data byte in the structure following; The ID is byte #3 . Byte #1 is the LSB of the ID Byte #2 is the MSB of the ID Byte #3 is the first data byte of the structure.
size	The number of bytes contiguous bytes contained in the data (LSB to MSB). A negative number indicates signed data; a positive number indicates unsigned data.
multiplier	The conversion factor (optional)
format	The format specifies the number of decimal places to output (optional).

Additionally, if the line has a plus sign (+) at the end of the line, the addition is performed on the data in the line and the line following.

For example, to decode the range of beam 1 through 4 in the bottom track data the following lines should be included.

```
0600,17,2,0.01,0,+      ; Beam #1 Range (cm)
0600,78,1,655.36,2      ; Beam #1 Range MSB
0600,19,2,0.01,0,+      ; Beam #2 Range (cm)
0600,79,1,655.36,2      ; Beam #2 Range MSB
0600,21,2,0.01,0,+      ; Beam #3 Range (cm)
0600,80,1,655.36,2      ; Beam #3 Range MSB
0600,23,2,0.01,0,+      ; Beam #4 Range (cm)
0600,81,1,655.36,2      ; Beam #4 Range MSB
$L
```

In order to simply output of the profile data you can use a combination of two lines. For example, to decode the velocity the following lines should be included.

```
$P idB, offsetB, typeB, offset, bins
idP, offsetP, typeP, multiplier, formatP
```

Where

\$P	Output simplified profile data
idB, offsetB, typeB	points to the number that describes the number of bins (normally 000,10,1)
offset	is the number of bytes to advance to the next bins (velocity is 8 bytes, intensity is 4 bytes)
Bins	are the string that describes what bins to output. For example, "1-10" would output bins one through 10, "1,5,7" would output bins one, five and seven, or "1-5,10-20" would output bins one through five and ten through twenty.
idP, offsetP, typeP	points to the data in the first bin for the profile (0100,3,-2 in the example shown below)

```
0080,3,2      ; Ensemble Number
$P0000,10,1,8,"1-255"      ; Starting Profile
0100,3,-2      ; Beam 1 Data
$L
$L
```



NOTE. See the included decoder files (*.DEC) for examples. To create your own decoder file, simply cut and paste the appropriate offset information into your own ASCII decoder file. The figures in the Command and Output Data Format guide should aid in creating decoding files.

Table 5: Decoder Files Included with RDI Tools CD

DEC File Name	Description
ADC.DEC	Decoder file for only the ADC channel data
BOTTOM.DEC	Bottom Track Data Decoder File
BTNMACOG.DEC	NMEA Bottom Track Data Decoder Information
BTVEL.DEC	Bottom Track velocity Data Decoder File
DEPTH.DEC	Decoder file for only the Transducer Depth data

Continued Next Page

Table 5: Decoder Files Included with RDI Tools CD (continued)

DEC File Name	Description
DISBIN1.DEC	Decoder file for only the Bin 1 Distance
ENSNO.DEC	Decoder file for only the Ensemble Number data
ENSTIME.DEC	Decoder file for only the Ensemble time Real-Time Clock
ENSTIME2.DEC	Decoder file for Ensemble Number and Time Decoder file when collecting both BB/NB OS Ensembles
EX.DEC	Decoder file for only the Coordinate Transformation (EX)
FIXDLDR2.DEC	Fixed Leader decoder file for OS ensembles with BB/NB pings
FIXEDLDR.DEC	Binary Fixed Leader Decode File
H.DEC	Decoder file for only the Heading data
HDR1.DEC	Header Decoder File ID 7F7F
HDR2.DEC	Header Decoder File ID 0000
HPR.DEC	Binary Fixed & Variable Leader Decode File for Heading, Pitch, and Roll
ID.DEC	Decoder File for decoding the ID numbers
ID2.DEC	Decoder File for decoding the ID numbers from OS Ensembles with both BB/NB Pings
IDSTNDRD.DEC	Standard ID Number Decoder File
LATLONG.DEC	Decoder file used to extract the Lat and Long values from the VMDAS ENS, ENX, STA, and LTA data files
LDR&NMEA.DEC	Variable Leader and NMEA Heading Data Decoder Information
LEADER.DEC	Binary Fixed & Variable Leader Decode File
LEADER2.DEC	Binary Fixed & Variable Leader Decoder File for NB Data when Collecting OS ensembles with both BB and NB pings
NGOOD.DEC	Decoder file for only the Number good pings (25 bins only)
NMEA.DEC	Decoder file for removing NMEA data from files created by VMDAS
NMEA_ENS.DEC	NMEA Data Decoder Information Ensemble Number
NMEAVEL.DEC	NMEA Data Decoder Information Velocity
SALIN.DEC	Decoder file for only the Salinity data
SUMSQR.DEC	Decoder file for only the sum of (velocity^2) (25 bins only)
SUMVEL.DEC	Decoder file for only the velocity sum (25 bins only)
TIME.DEC	Decoder file for removing Date and Time from Ensemble Data
UPDOWN.DEC	Decoder file for only the Orientation data
WBCMD.DEC	Decoder file for only the Fixed Leader WB Command setting Decoder Information
WHCPUsn.dec	Binary Fixed Leader Decode File for WorkHorse CPU Serial Number
WINRIVGPS.DEC	WINRIVER Navigation Decode File



NOTE. The LEADER.DEC is the master ASCII Decoder file for variable and fixed leader data. Use the BOTTOM.DEC file for only the Bottom Track data.

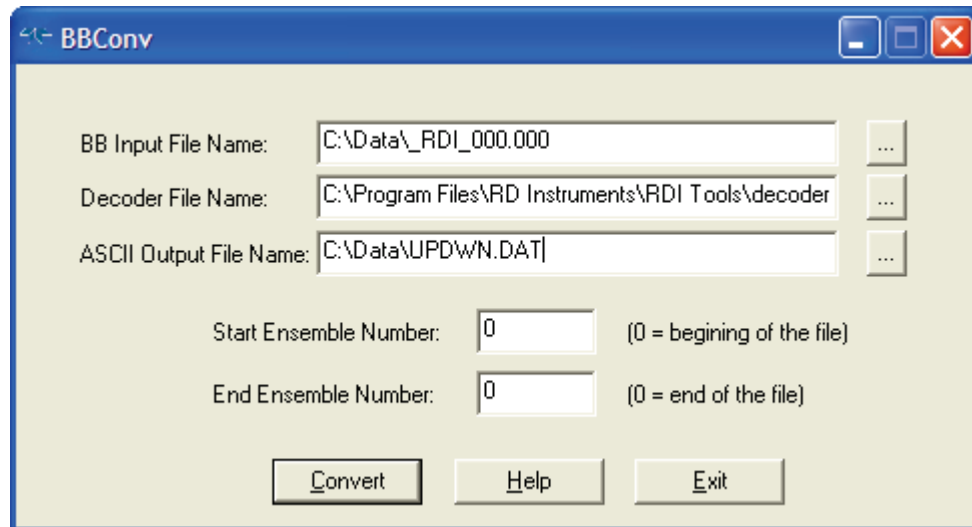
5.1 Example Using UPDWN.DEC

Teledyne RD Instruments' software only reads the first ensemble to determine the orientation of the ADCP. If your system started pinging and recording data before it was in the correct orientation, then the programs will display the data upside down. If you collected data in beam coordinates, our software will also display the velocity components with the wrong sign. To correct this you must convert the configuration byte in the ADCP leader data containing the orientation setting to the correct value. The following paragraphs explain how to use *BBConv* and *BBMerge* to convert the configuration byte in the ADCP data set to all up or all down.

- a. You must first determine which ensemble number has the proper setting of the configuration byte. Use the *BBLIST* program to view your ADCP data. Once your data is loaded press <ALT+D> and then **A** to enter the ADCP setup screen in *BBLIST*. Look at the setting in the **ADCP Hardware and Firmware** window for the ADCP orientation.

Now, press the + key until you move to the ensemble that has the correct orientation for your deployment. Record this ensemble number for future use.

- b. Use *BBConv* to copy the configuration byte from the ADCP binary data file to an ASCII data file. For this example, we will assume that your raw data file is *_RDI_000.000*.



Example UPDWN.DEC file:

```
; Binary Fixed & Variable Leader Decode File
0080,3,2           ; Ensemble Number
0000,5,2           ; System Configuration
$L
```

An ASCII text file called *UPDOWN.DAT* will be created. It will have two columns of data. The first column will be the ensemble number and the second column will be the configuration byte.



NOTE. *BBConv* will automatically go to the next extension (i.e. *_RDI_000.001*) if it exists.

- c. Use a text editor to edit the *UPDOWN.DAT* file so that the configuration byte is the same for all ensembles. Locate the ensemble number you recorded in Step “a” and copy the configuration byte setting to all of the other ensembles. Most text editors will allow you to use a “find and replace” option.

After all of the configuration bytes have been changed, save the file. Be sure to not add any other characters to the file.

- d. Now use *BBMerge* to write the new configuration byte back into your data set. *BBMerge* will write the data into a *new* data file. It will not affect the original raw data file.

BBMerge

BB Input File Name: C:\Data_RDI_000.000

Decoder File Name: es\RD Instruments\RDI Tools\decoders\UPDOWN.DEC

ASCII File Name: C:\Data\UPDOWN.DAT

BB Output File Name: C:\Data\NEW.000

Start Ensemble Number: 0 (0 = beginning of the file)

End Ensemble Number: 0 (0 = end of the file)

Merge Help Exit



NOTE. *BBMerge* will automatically go to the next extension (i.e. *_RDI_000.001*) if it exists.

- e. You can now playback your data and the orientation will be correct. If you collected data in beam coordinates, the velocity data will now have the correct signs.

5.2 Example Using ADC.DEC

You can use *BBConv* to obtain the ADC channel information and the heading, pitch, roll, standard deviation information. Teledyne RD Instruments' *BBLIST* software does not allow you to output the ADC data or the heading, pitch, and roll standard deviation information. Using the *BBConv* program it is possible to output this information into an ASCII text file.

- a. You must use the *BBConv* program to copy the ADC, heading, pitch, and roll bytes from the WorkHorse or BroadBand ADCP binary data file to an ASCII data file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *TEST.000*.

BB Input File Name	TEST.000
Decoder File Name	ADC.DEC
ASCII Output File Name	ADC.DAT

An ASCII text file called *ADC.DAT* will be created. It will have 2 columns of data. The first column will be the ensemble number and the second column will be the heading bytes.



NOTE. *BBCONV* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

5.3 Example Using DEPTH.DEC

Use *DEPTH.DEC* to change the depth in the BroadBand, WorkHorse, or Ocean Surveyor ADCP leader data. Teledyne RD Instruments' software only reads the first ensemble to determine the depth of the ADCP. If you entered the wrong depth during the deployment setup of your system then our software programs will set the depth of the bins incorrectly.

To correct this you must change the depth in the variable leader data for at least the first ensemble. The following explains how to use *BBConv* and *BBMerge* to change the depth bytes in the WorkHorse and BroadBand ADCP data set to a new value.

- b. You must use the *BBConv* program to copy the depth bytes from the WorkHorse or BroadBand ADCP binary data file to an ASCII data file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *LEG3009_000000.ENR*.

BB Input File Name	LEG3009_000000.ENR
Decoder File Name	DEPTH.DEC
ASCII Output File Name	DEPTH.DAT

- c. An ASCII text file called *DEPTH.DAT* will be created. It will have two columns of data. The first column will be the ensemble number and the second column will be the depth byte.



NOTE. *BBConv* will automatically go to the next file increment (i.e. *LEG3009_000001.ENR*) if it exists.

- d. You must now edit the *DEPTH.DAT* file so that the depth of at least the first ensemble is the correct depth. You may edit the depth for all ensembles but only the first one needs to be edited for our software to scale your depth cells correctly. Locate ensemble number one and change it to match the correct depth.
- e. If you wish to change all ensembles then most text editors will allow you to use a find and replacement option. After the depth bytes have been changed save the file. Be sure to not add any other characters to the file.
- f. You can now use *BBMerge* to write the new depth bytes back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	LEG3009_000000.ENR
Decoder File Name	DEPTH.DEC
ASCII Output File Name	DEPTH.DAT
BroadBand Output File Name	NEW.000

The file *NEW.000* will be the corrected data file.



NOTE. *BBMerge* will automatically go to the file increment (i.e. *LEG3009_000001.ENR*) if it exists.

- g. You can now playback your data and the depth of your ADCP and your bins will be correct.

5.4 Example Using ENSEMBLE.DEC

Use *BBConv* and *BBMerge* to change the ensemble number in the Work-Horse or BroadBand ADCP leader data. Teledyne RD Instruments' software reads the ensemble number in each ensemble to determine the sequence of the ensembles. If you stop and start a recording cycle in the ADCP the ensemble number will re-initialize back to one. Some programs are sensitive to this (TRDI's *WinADCP* program for example) and will not combine the files together later.

To correct this you must change the ensemble number in the files following your first file so that they continue the ensemble number sequence. The following explains how to use *BBConv* and *BBMerge* to change the ensemble number bytes in the WorkHorse and BroadBand ADCP data sets to new values.

- a. You must use the *BBConv* program to copy the ensemble number bytes from the WorkHorse or BroadBand ADCP binary data file to an ASCII data file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *TEST.000*.

BB Input File Name	TEST.000
Decoder File Name	ENSNO.DEC
ASCII Output File Name	ENSNO.DAT

- b. An ASCII text file called *ENSNO.DAT* will be created. It will have a single column of data. This column will be the ensemble number in the data set you recorded.



NOTE. *BBConv* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- c. You must now edit the *ENSNO.DAT* file so that the ensemble numbers are sequential. Note, the ensemble number must roll over at ensemble 65535. Most spreadsheet programs would allow you to do this editing very easily. After the ensemble numbers have been changed save the file. Be sure to not add any other characters to the file.
- d. You can now use *BBMerge* to write the new ensemble number bytes back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	TEST.000
Decoder File Name	ENSNO.DEC
ASCII Output File Name	ENSNO.DAT
BroadBand Output File Name	NEW.000

The file *NEW.000* will be the corrected data file.



NOTE. *BBMerge* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- e. You can now playback your data and the ensemble numbers will be sequential.

5.5 Example Using EX.DEC

Use *BBConv* and *BBMerge* to set the EX command setting in the BroadBand ADCP leader data. Teledyne RD Instruments' software reads the EX Command setting to determine the coordinate system the ADCP was set in. If your leader data recorded the wrong coordinate system then you will not be able to use TRDI's software to correctly display the data in the coordinate system you wish.

To correct this you must convert the EX Command byte in the BroadBand ADCP leader data to the correct value. The following explains how to use *BBConv* and *BBMerge* to convert the configuration byte in the BroadBand ADCP data change the setting to correct value.

- a. You must use the *BBConv* program to copy the EX Command byte from the BroadBand ADCP binary data file to an ASCII data file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *TEST.000*.

BB Input File Name	TEST.000
Decoder File Name	EX.DEC
ASCII Output File Name	EX1.TXT

- b. An ASCII text file called *EX1.TXT* will be created. It will have two to three columns of data. The first column will be the EX Command setting, the second column will be the ensemble number LSB of the ensemble number and the third column will be the MSB of the ensemble number.



NOTE. *BBConv* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- c. You must now edit the *EX1.TXT* file so that the EX Command byte is the same and correct for all ensembles. Change the EX Command byte from its present value to the values shown here for your correct coordinate system:

0	= Beam
15	= Instrument
23	= Ship
31	= Earth

After all of the configuration bytes have been changed save the file to a new file called *EX2.TXT*. Be sure to not add any other characters to the file.

- d. You can now use *BBMerge* to write the new configuration byte back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	TEST.000
Decoder File Name	EX.DEC
ASCII Output File Name	EX2.TXT
BroadBand Output File Name	NEW1.000

The file *NEW1.000* will be the corrected data file.



NOTE. *BBMerge* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- e. You can now playback your data and the coordinate system will be correct.

5.6 Example Using H.DEC

Use *H.DEC* to change the heading in the WorkHorse or BroadBand ADCP leader data. Teledyne RD Instruments' software reads the heading in each ensemble to determine the direction the ADCP is pointing. If you entered the wrong heading or your heading is incorrect during the deployment of your system then our software programs will calculate the wrong direction of the velocity data.

To correct this you must change the heading in the variable leader data for at each ensemble. The following explains how to use *BBConv* and *BBMerge* to change the heading bytes in the WorkHorse and BroadBand ADCP data set to a new value.

- a. You must use the *BBConv* program to copy the heading bytes from the WorkHorse or BroadBand ADCP binary data file to an ASCII data file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *TEST.000*.

BB Input File Name	TEST.000
Decoder File Name	H.DEC
ASCII Output File Name	HDG.DAT

- b. An ASCII text file called *HDG.DAT* will be created. It will have two columns of data. The first column will be the ensemble number and the second column will be the heading bytes.



NOTE. *BBConv* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- c. You must now edit the *HDG.DAT* file so that the heading of each ensemble is correct. Most text editors will allow you to use a find and replacement option to change a specific value.
- d. After the heading bytes have been changed save the file. Be sure to not add any other characters to the file.
- e. You can now use *BBMerge* to write the new heading bytes back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	TEST.000
Decoder File Name	H.DEC
ASCII Output File Name	HDG.DAT
BroadBand Output File Name	NEW.000

The file *NEW.000* will be the corrected data file.



NOTE. *BBMerge* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- f. You can now playback your data and the heading of your ADCP and the direction of your velocity data will be correct.

5.7 Example Using NBID2BB.DEC

Use *NBID2BB.DEC* to change the NarrowBand ID number to the BroadBand ID number. The Ocean Surveyor ADCP is capable of collecting both BroadBand and NarrowBand pings in the same ensemble. When this happens it writes the BroadBand data (fixed leader, variable leader, velocity, correlation, echo intensity, and percent good) to the standard ID numbers (0000, 0080, 0100, 0200, 0300, and 0400 respectively). The NarrowBand data (fixed leader, variable leader, velocity, correlation, echo intensity, and percent good) is given new ID numbers (0001, 0081, 0101, 0201, 0301, and 0401 respectively). This is done so that 2 different data types can exist in the same ensemble.

Teledyne RD Instruments' software reads only the standard the ID numbers when decoding binary data. As a result only the BroadBand ensembles will be decoded. By changing the ID numbers of the both the BroadBand and NarrowBand data types it is possible to have the NarrowBand data be read by either *WinADCP* or *VmDas*.



NOTE. Currently the *BBLIST* program will continue to read the BroadBand data fixed leader type but does appear to display the velocity data from the NarrowBand ensembles once this procedure has been followed.

- a. You must use the *BBConv* program to copy the ID numbers from both the BroadBand and NarrowBand data types into a text file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *LEG3009_000000.ENR*.

BB Input File Name	LEG3009_000000.ENR
Decoder File Name	ID2.DEC
ASCII Output File Name	1BBNBID.TXT

An ASCII text file called *1BBNBID* will be created. It will have 14 columns of data. The first two columns will be the ensemble number LSB and MSB. Columns 3 through 8 will be the ID numbers for the BB data and columns 9 through 14 will be the ID numbers for the NB data.



NOTE. *BBConv* will automatically go to the next file increment (i.e. *LEG3009_000001.ENR*) if it exists.

- b. You must now edit the *1BBNBID.TXT* file so that the ID numbers for the BroadBand data are changed from 0, 128, 256, 512, 768, 1024 to 2, 130, 258, 514, 770, 1026 respectively.



NOTE. The values in the file and shown here are the decimal equivalent of the hex values actually found in the binary data files.

You must also change the ID numbers for the NarrowBand data from 1, 129, 257, 513, 769, 1029 to 0, 128, 256, 512, 768, 1024 respectively.

While most text editors will allow you to use a find and replace option to change a specific value, it is easier to load this data into a spreadsheet program such EXCEL and then replace the values for the first row of data and then copy that through the rest of the ensembles. If done through *EXCEL* the changed file can be saved as a MS-DOS CSV file. A CSV file will contain all of the comma delimiters required for use with *BBMerge* in the next step.



NOTE. You will have to change the file name to *.TXT. It is best to save your modified data to a different file name such as *2BBNBID.TXT*.

When working with this file be sure to not add any other characters to the file as this cause incorrect data to be merged into your data set.

- c. You can now use *BBMerge* to write the new ID Number bytes back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	LEG3009_000000.ENR
Decoder File Name	ID2.DEC
ASCII Output File Name	2BBNBID.TXT
BroadBand Output File Name	NEW39_000000. ENR

The file *NEW39ENR.000* will be the corrected data file and will contain the ID numbering.



NOTE. *BBMerge* will automatically go to the next extension (i.e. *LEG3009_000000.ENR*) if it exists.

- d. You can now reprocess the raw data with *VmDas* so that you are using the NarrowBand data.

5.8 Example Using NMEA.DEC

Use *NMEA.DEC* to extract the NMEA data from the *VmDas* STA and LTA files. Teledyne RD Instruments' software program *VmDas* creates Short Term Average (STA) and Long Term Average (LTA) files. These files have the exact same format as the original output of your ADCP, but contain one more data type. This data type is called NMEA Data. It contains the latitude, longitude, heading, pitch, and roll information that was collected when using *VmDas*.

The following procedure explains how to use *BBConv* to extract the NMEA data from the STA and LTA files.

- a. You must use the *BBConv* program to extract the heading bytes from the STA or LTA *VmDas* binary data files into an ASCII data file. Run *BBConv*. For this example, we will assume that your file name is *ADCP007_000000.STA*.

BB Input File Name	ADCP007_000000.STA
Decoder File Name	NMEA.DEC
ASCII Output File Name	ADCP007.TXT



NOTE. *BBConv* will automatically go to the next file increment (i.e. *ADCP007_000001.STA*) if it exists.

An ASCII text file called *ADCP007.TXT* will be created. It will contain 32 columns of data with each column being comma delimited. The

meaning of each column is explained in the NMEA section of the STA and LTA Data Format description or you can refer to the NMEA.DEC text file.

- b. Several of the fields in the ASCII file will have to be converted or scaled into their proper units. To scale the data refer to the NMEA section of the STA and LTA Data Format description.

Included with this is an example *Excel* spread sheet, *NMEA.XLS* that has the formulas for scaling all of the data. To use this spreadsheet, open your *BBConv* output text file with *Excel* and then copy this data onto the **BBConv DATA IN** tab in the *NMEA.XLS* spread sheet. You can then click on the **REAL UNITS** tab and you find your data has been converted.



NOTE. An example of data from the file *ADCP007.TXT* is already included in the *NMEA.XLS* spreadsheet. You can copy your data right over the data already in the file or copy the *NMEA.XLS* file to another name. Keep in mind that, the **Real Units** tab is only setup for a limited number of averages. Therefore, you may need to copy the formulas from the final row to additional rows.

5.9 Example Using SALIN.DEC

Use *SALIN.DEC* to change the salinity in the WorkHorse or BroadBand ADCP leader data. Teledyne RD Instruments' software reads the salinity in each ensemble to determine the speed of sound for the ADCP velocity measurement. If you entered the wrong salinity during the deployment setup of your system then our software programs will calculate the velocity and depth of each bin incorrectly.

To correct this you must change the salinity in the variable leader data for at each ensemble. The following explains how to use *BBConv* and *BBMerge* to change the salinity bytes in the WorkHorse and BroadBand ADCP data set to a new value.

- a. You must use the *BBConv* program to copy the salinity bytes from the WorkHorse or BroadBand ADCP binary data file to an ASCII data file. To do this, use the following example of what to enter at the *BBConv* screen. For this example, we will assume that your file name is *TEST.000*.

BB Input File Name	TEST.000
Decoder File Name	SALIN.DEC
ASCII Output File Name	SALIN.DAT

- b. An ASCII text file called *SALIN.DAT* will be created. It will have two columns of data. The first column will be the ensemble number and the second column will be the salinity byte.



NOTE. *BBConv* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- c. You must now edit the *SALIN.DAT* file so that the salinity of each ensemble is correct. Most text editors will allow you to use a find and replacement option to change a specific value. After the salinity bytes have been changed save the file. Be sure to not add any other characters to the file.
- d. You can now use *BBMerge* to write the new salinity bytes back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	TEST.000
Decoder File Name	SALIN.DEC
ASCII Output File Name	SALIN.DAT
BroadBand Output File Name	NEW.000

The file *NEW.000* will be the corrected data file.



NOTE. *BBMerge* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- e. You can now playback your data and the salinity of your ADCP and your bins will be correct.

5.10 Example Using SUM.DEC

Use *SUM.DEC* to pull out the sum of square data. Teledyne RD Instruments' recently added the following data to its data format: Number of good pings, Sum of Velocity Squared, and Sum of Velocities. With this information it is possible to calculate the standard deviation of the velocity data for each bin in each ADCP ensemble.

Regrettably, the standard TRDI software (such as *BBLIST* and *WinADCP*) does not know how to decode or use this data. In order to gain access to this data we created the *BBConv* program and the Decode files to produce ASCII files. These files can then be loaded into a spreadsheet to calculate the standard deviation.

The following explains how to use *BBConv* and the Decode files to convert the above-mentioned data into ASCII files.

- a. You must first modify each of the Decode files (NGOOD, SUMSQR, SUMVEL) that have been provided so that you are decoding all the bins of data you wish. Each of the files already will decode the first 25 bins of data in each ensemble.

To increase (or decrease) the numbers of bins use a text editor to modify each of the Decode files. You can copy the format that is already there and just change the pointer location value to increase the number of bins being decoded.

- b. You will need to run the *BBConv* program three separate times (once for each data type) on each data set that you have.

For example, if the data set was named *RDI001.000 - RDI001.099* and we wanted the Number of Good Pings then the format would be as follows:

BB Input File Name	RDI001.000
Decoder File Name	NGOOD.DEC
ASCII Output File Name	NGOOD.TXT



NOTE. *BBConv* will automatically go to the next extension (i.e. *RDI001.001*) if it exists.

This will cause the program to go through the entire data set from extension .000 through .099 and output the number of good pings for each bin in a text file called *NGOOD.TXT*. You can choose any name for the file you wish. You would then repeat the above for the Sum of Velocity Squared and Sum of Velocities, i.e.:

BB Input File Name	RDI001.000
Decoder File Name	SUMSQR.DEC
ASCII Output File Name	SUMSQR.TXT

and

BB Input File Name	RDI001.000
Decoder File Name	SUMVEL.DEC
ASCII Output File Name	SUMVEL.TXT

5.11 Example Using TIME.DEC

The following procedure explains how to use *TIME.DEC* to extract the Date and Time data from the BroadBand, WorkHorse, or Ocean Surveyor ADCP data. Teledyne RD Instruments' software stamps each ensemble with the date and time. If the date or time has been set wrong for your deployment it is possible to change the date or time using the following procedure.

- a. You must use the *BBConv* program to extract the date and time bytes from the variable leader data for your data set. Run *BBConv*. For this example, we will assume that your file name is *ADCP001_000000.STA*.

BB Input File Name	ADCP001_000000.STA
Decoder File Name	TIME.DEC
ASCII Output File Name	ADCP0000.TXT



NOTE. *BBConv* will automatically go to the next file increment (i.e. *ADCP001_000001.STA*) if it exists.

An ASCII text file called *ADCP0000.TXT* will be created. It will contain seven columns (fields) of data with each column containing two characters (one byte) of data. Each field is comma delimited. The meaning of each column is as follows: YY, MM, DD, HH, MM, SS, HH

Where: YY = Year, MM = Month, DD = Day, HH = Hour, MM = Minute, SS = Second, HH = Hundredths of Seconds

- b. You must now edit the *ADCP0000.TXT* file so that the Date and Time of each ensemble is correct. Most text editors will allow you to use a find and replacement option to change a specific value.

After the date and time bytes have been changed save the file. Be sure to not add any other characters to the file.

- c. You can now use *BBMerge* to write the new date and time bytes back into your data set. *BBMerge* will actually write the data into a new data file. It will not affect the original file. To do this, use the following example of what to enter at the *BBMerge* screen.

BB Input File Name	ADCP001_000000.STA
Decoder File Name	TIME.DEC
ASCII Output File Name	ADCP0000.TXT
BroadBand Output File Name	NEW0000.000

The file *NEW0000.000* will be the corrected data file.



NOTE. *BBMerge* will automatically go to the next extension (i.e. *NEW0000.001*) if it exists.

- d. You can now playback your data and the heading of your ADCP and the date and times of your data (*NEW0000.**) will be correct.

5.12 Example Using WHCPUsn.DEC

Use *WHCPUsn.DEC* to obtain the WorkHorse CPU serial number. Teledyne RD Instruments' (TRDI) WorkHorse ADCPs all are given unique serial numbers for both the system and the individual boards within a system. The system serial number is selected by TRDI and is provided on the TRDI logo and sticker on the pressure case. The individual board serial numbers are electronic serial numbers programmed at TRDI. The overall system serial number is not logged in the data collected by the WorkHorse ADCP. However, the WorkHorse CPU electronic serial number is logged in all WorkHorse data sets as part of the fixed leader data of all WorkHorse ADCPs with firmware version 8.20 and higher.

The WorkHorse electronic serial number can also be viewed when the PS0 command is sent to the ADCP. A copy of the PS0 command is stored in all TRDI testing files that are run as part of either TRDI's DOS based programs or TRDI's Windows program *BBTalk* and the associated script file for testing.

Using the WorkHorse CPU electronic serial number recorded in these tests it is possible to cross-reference this to your system serial number. However, TRDI recommends that you create your own log of electronic and system serial numbers, because you may update or repair your WorkHorse system in the field and this would change the electronic serial number.

The following procedure explains how to read out the WH CPU electronic serial number from your data set so that you can confirm which system the data set is from. Please note you must have created a cross-reference list first to make this confirmation. If you have not, then it is possible for TRDI to perform this cross-reference assuming that you have not changed the WorkHorse CPU board in the field.

- a. You must use the *BBConv* program to copy the CPU electronic serial number from the WorkHorse ADCP binary data file to an ASCII data file. Run *BBConv*. For this example, we will assume that your file name is *TEST.000*.

BB Input File Name	TEST.000
Decoder File Name	WHCPUSN.DEC
ASCII Output File Name	WHCPUSN.TXT



NOTE. *BBConv* will automatically go to the next extension (i.e. *TEST.001*) if it exists.

- b. An ASCII text file called *WHCPUSN.TXT* will be created. It will have eight columns of data (each column separated by a comma). Each row will repeat the same eight values, as this is the same CPU serial number

recorded in each ensemble. Therefore, only the first row of numbers is required.

To match the values in the first row with the CPU serial number you must convert each comma-separated value from its decimal value to a hexadecimal value.

- c. Use the CPU serial number you have obtained and your PS0 results to cross reference to determine your WorkHorse ADCP system serial number. The following example illustrated what checks would be done.

1. Example of PS0 Results from a WorkHorse ADCP s/n 100

```
[BREAK Wakeup A]
WorkHorse Broadband ADCP Version 16.17
Teledyne RD Instruments (c) 1996-2001
All Rights Reserved.
>ps0
Instrument S/N: 0
Frequency: 307200 HZ
Configuration: 4 BEAM, JANUS
Match Layer: 10
Beam Angle: 20 DEGREES
Beam Pattern: CONVEX
Orientation: DOWN
Sensor(s): HEADING TILT 1 TILT 2 TEMPERATURE
Temp Sens Offset: 0.00 degrees C

CPU Firmware: 16.17 [0]
Boot Code Ver: Required: 1.13 Actual: 1.13
DEMOM #1 Ver: ad48, Type: 1f
DEMOM #2 Ver: ad48, Type: 1f
PWRTIMG Ver: 85d3, Type: 5

Board Serial Number Data:
34 00 00 02 67 AD 9A 09 CPU727-2000-00H (first 16 values are the
CPU serial number)
41 00 00 00 09 A0 C6 09 PIO727-3000-00B
3B 00 00 00 62 72 DD 09 DSP727-2001-04D
DF 00 00 00 0E 1A 46 09 REC727-1000-04A
>
```

2. Example of the contents of *WHCPUSN.TXT* (the capture of the output from *BBCnv*)

```
52,0,0,2,103,173,154,9
52,0,0,2,103,173,154,9
52,0,0,2,103,173,154,9
52,0,0,2,103,173,154,9
52,0,0,2,103,173,154,9
....repeating
```

3. Converting the first row of values above from decimal to hexadecimal you will obtain

```
52,0,0,2,103,173,154,9 --> 34,0,0,2,67,AD,9A,9
```

4. The converted values match the first 16 digits of the CPU line in the PS0 output.

6 Software History

6.1 BBTalk History

Version 3.08 (07/09)

- Fixed issues recovering loop recorder from Channel Master.
- Added RiverRay compatibility.
- Fixed \$COM command issues.
- Fixed BBTalk crashing if double-clicked on empty window.

Version v3.07 (05/08)

- Changed logo, icon, company name, and address.
- Fixed problem with Bluetooth drivers.
- Change label under Transfer menu from "PC Time" to "PC Time to ADCP".
- Added "Wait For Prompt in Script File" option.
- Fixed problems with launching script files from Windows Explorer.
- Fixed problem with file extension of the log file when using script.
- Fixed problem with changing baud rate through script file.
- Changed loop recovery progress bar behavior to be based on total capacity instead of the amount of recorded data.
- Fixed problem with recovering loop recorder data if Channel Master has gone to sleep.
- Disabled "Recover Recorder" menu item for Broadband.
- Added AccQmin and DVS to the device dropdown menu.
- Fixed problem with scripts to restore WH system calibration.
- Added Vista compatibility.

Version 3.06 (03/05)

- Fixed the ability to send script files to the currently opened port by clicking on a script file icon.

Version 3.05 (Internal Release Only)

- Added option to disable screen output during data recovery.

- Fixed the screening of CB011 command for Workhorse and Channel Master.
- Fixed the problem with aborting **Recorder Recovery** process.
- Fixed the screening of MY command.
- Removed invalid commands from OS2 sample script file.
- Added option to echo typed characters.
- Fixed **AutoDetect** feature when used with soft break.
- Changed error message when Recorder not detected.
- Added a warning message at the end of the program if ADCP is pinging.
- Fixed the screening of CB command.
- Added **NEMO** selection in **Connect To** dialog.
- Added baud rate 300 as a selection for Workhorse.
- Fixed error handling for corrupted script files.

Version 3.04 (11/03)

- Fixed problem with sending BBxxx command.
- Added support for Channel Master loop-recorder.

Version 3.03 (05/03)

- Added a display of the log file name.
- Changed program icon.
- Changed “Break” icon when software break is selected.

Version 3.02 (11/02)

- Added support for the software break (“===”) for radio modems.
- Changed “...” button to “Browse...”.
- Added support for ADFM that does not have RR-command. Program now detects if user types RY-command. If after RY-command the user presses Ctrl+PgDN than the program goes directly to Y-modem protocol without sending a break and the RR command.
- Fixed the problem of hanging up when changing baud rate during pinging.

- Added ability to send PC time to the instrument.
- Added ability to send bytes in HEX form and script command "\$S" to send binary data (in hex format).
- Added "\$A" to switch to ASCII display. "\$H" switches to HEX display.
- Automatically tile windows under opening.
- Changed the way list of available ports is detected.
- Fixed problem with accelerator keys when running script.
- Added "\$F" to script to find ADCP.
- Added "\$T" to set ADCP clock to PC clock.
- Added menu item to call help.
- Added "\$t" to send any text and CR/LF at the end.
- Added "\$D" or "\$d" to accept decimal seconds.
- Fixed the problem with program crashing when running a script and closing a window.
- Fixed problem with sending command in binary mode for Narrowband.
- Fixed the problem with listing all the files if two memory cards were present.
- Fixed problem with redisplaying during fast pinging.
- Fixed problem with opening ports > COM9.
- Fixed problem with program crashing after some programs used a port.
- Improved a way the program remembers opened ports.
- Fixed problem with handling higher COM ports (COM13 and up)

Version 3.01 (12/01)

- Initial release

6.2 BBCheck History

Version 2.04 (05/08)

- Changed logo, icon, company name, and address.
- Added support for Waves data files (packets).
- Added more file information to the log file.
- Added Vista compatibility.

Version 2.03 (11/03)

- Added ensemble size and data structure length to the log file.
- Improved selection of the log file path.

Version 2.02 (05/03)

- Changed program icon.
- Fixed the problem with help button.

Version 2.01 (11/02)

- Changed IDs description for Beam 5 data.
- Fixed the problem that was caused by the file that had a lot of garbage characters. Program would stop searching for the next ensemble after 65535*2 bytes (would seem enough under normal situation). Now after the first ensemble is found the search will continue until the end of the file.
- Limited ensemble size to 1 MB and rechecked for new ensemble if end of file reached.

Version 2.00 (12/01)

- First release.

6.3 BBConv History

Version v2.03 (05/08)

- Changed logo, icon, company name, and address.
- Corrected few decoder files.
- Added Vista compatibility.

Version 2.02 (05/03)

- Changed program icon.
- Fixed the problem with help button.

Version 2.01 (11/02)

- Fixed the following problem: if an ID number is requested to be decoded and it does not exist in the data file then no data would be output for any ID numbers that are listed within the same string. A string is defined as a list of ID numbers without a line feed being placed between them.
- Fixed the problem that was caused by the file that had a lot of garbage characters. Program would stop searching for the next ensemble after 65535*2 bytes (would seem enough under normal situation). Now after the first ensemble is found the search will continue until the end of the file.
- Limited ensemble size to 1 MB and rechecked for new ensemble if end of file reached.

Version 2.00 (12/01)

- First release.

6.4 BBMerge History

Version 2.03 (05/08)

- Changed logo, icon, company name, and address.
- Added Vista compatibility.

Version 2.02 (05/03)

- Changed program icon.
- Fixed the problem with help button.

Version 2.01 (11/02)

- Fixed the problem that was caused by the file that had a lot of garbage characters. Program would stop searching for the next ensemble after 65535*2 bytes (would seem enough under normal situation). Now after the first ensemble is found the search will continue until the end of the file.
- Limited ensemble size to 1 MB and rechecked for new ensemble if end of file reached.

Version 2.00 (12/01)

- First release.

6.5 BBSlice History

Version 2.03 (05/08)

- Changed logo, icon, company name, and address.
- Added Vista compatibility.

Version 2.02 (05/03)

- Changed program icon.
- Fixed the problem with help button.

Version 2.01 (11/02)

- Fixed the problem that was caused by the file that had a lot of garbage characters. Program would stop searching for the next ensemble after 65535*2 bytes (would seem enough under normal situation). Now after the first ensemble is found the search will continue until the end of the file.
- Limited ensemble size to 1 MB and rechecked for new ensemble if end of file reached.

Version 2.00 (12/01)

- First release.

6.6 BBss History

Version 2.02 (05/08)

- Changed logo, icon, company name, and address.
- Added Vista compatibility.

Version 2.01 (05/03)

- Changed program icon.
- Fixed the problem with help button.

Version 2.00 (12/01)

- First release.

6.7 BBSub History

Version 2.06 (11/09)

- Added a filter to change 3 beams to 4 beams in fixed leader.

Version 2.05 (07/09)

- Added Filter option for running BBSub from Command line.

Version 2.04 (05/08)

- Changed logo, icon, company name, and address.

- Renamed option to extract OS narrowband data.
- Added Vista compatibility.

Version 2.03 (11/03)

- Implemented filter to extract OS Narrowband data.
- Implemented filter to force up-down looking sensor to be down-looking.

Version 2.02 (05/03)

- Added a filter to allow user to separate up-looking and down-looking data.
- Changed program icon.
- Fixed the problem with help button.
- Added filter to redo the ensemble numbers and recalculate checksum.
- Added ability to resample data.

Version 2.01 (11/02)

- Fixed the problem that was caused by the file that had a lot of garbage characters. Program would stop searching for the next ensemble after 65535*2 bytes (would seem enough under normal situation). Now after the first ensemble is found the search will continue until the end of the file.
- Limited ensemble size to 1 MB and rechecked for new ensemble if end of file reached.

Version 2.00 (12/01)

- First release.

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